

Nonlinear Aggregation of Local Likelihood Scores for Simulation-Based Inference

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Abstract

In many applications, the joint likelihood is intractable even though marginal, pairwise, or block likelihoods can be evaluated. We propose to use the corresponding local likelihood scores as inputs to simulation-based inference. The proposed architecture first transforms the local scores, then pools them within groups for which permutation invariance is appropriate. Identity Local-Score Aggregation (ILSA) pools the scores without a learned local feature, whereas Nonlinear Local-Score Aggregation (NLSA) appends a learned nonlinear feature to each score before pooling. The resulting score function is estimated without evaluating the joint likelihood by a tube version of Fisher-score matching. A neural posterior estimator is then conditioned on both a pilot estimate and the estimated score evaluated at the pilot. In two controlled examples, NLSA reduces the standardized mean squared error of the estimated full score by 40.6% and 32.1%, respectively, and also gives more accurate posterior approximations. In a Smith max-stable model with 79 sites, it reduces the integrated squared error of the estimated extremal-dependence surface by 14.0–18.2% at three parameter values.

Keywords: Local likelihood scores; Fisher-score matching; structured representations; simulation-based inference.

1 Introduction

In many applications, complex dependence or latent structure makes the joint likelihood difficult to evaluate. Examples include spatial extremes, elliptical random fields, latent-variable models, and Markov random fields. For these models it is often straightforward to simulate data, and marginal, pairwise, or block likelihoods may also be available even when the joint density is not [Smith and Stephenson, 2009, Varin and Vidoni, 2008]. Generic simulation-based inference (SBI) encoders can be used on the raw data, but must learn the complex dependencies, including these local structures, from simulations. Existing structured score-learning methods exploit an exact additive or compositional decomposition of the likelihood or posterior distribution when the likelihood can be factorized exactly, for example over independent observations or Markov transitions [Jiang et al., 2026, Gloeckler et al., 2025]. However, such factorizations are not available for models considered here. Nevertheless, their tractable local likelihoods already provide informative parameter gradients encoding marginal behavior, local dependence, lag or distance effects, and other scientific structure before any neural representation is learned.

Our main question is how to exploit this existing partial model geometry to enable more efficient statistical inference. Naively replacing the raw data with the collection of local scores in posterior conditioning seems merely to represent largely redundant information in a much higher-dimensional form; for example, the number of pairwise local scores is $O(n^2)$. Classical composite-likelihood methods combine local scores linearly with carefully chosen weights [Varin et al., 2011, Lindsay et al., 2011]; alternatively, posterior conditioning on multiple composite scores can automatically learn their combination [Li et al., 2026]. Let $g_0(X; \theta) := \nabla_{\theta} \log f(X | \theta)$ denote the Fisher score, where $f(X | \theta)$ is the likelihood. Such combinations can be efficient when the Fisher score is well represented by the linear span of the local scores, but their first-order information is restricted to that span [Varin et al., 2011, Lindsay et al., 2011, Li et al., 2026]. It is natural to provide a more flexible surrogate for the Fisher score by enlarging the function class formed by the local scores. The Fisher score is a natural target for Bayesian inference because, under standard local asymptotic conditions, the score at the true parameter is a locally information-preserving low-dimensional summary [Brehmer et al., 2020].

This work makes three main contributions to surrogate Fisher-score construction and amortized posterior inference. First, instead of applying a generic MLP to the concatenation of all local scores, we propose the architecture preserving within-group permutation invariance of homogeneous groups of local scores and having the same expressiveness as the generic MLP by apply shared within-group transformations to local scores. See Figure 1.

Second, we propose to learn a two-argument surrogate score surface $S(X; \beta)$ using Fisher-score matching in a tube area, and the study the choice of tube width which is the smoothing tuning parameter. Conditioning on the surrogate score function for posterior inference does not require structure of the scores like being unbiased estimating function or Bartlett identity which needs to be imposed when using for estimating-function based inference [Jiang et al., 2026].

Third, we propose to use the pair consisting of a pilot estimate of the data-generating parameter, $\hat{\theta}(X)$, and $S\{X; \hat{\theta}(X)\}$ as a summary vector for neural posterior estimation (NPE) in the sampling stage. This construction addresses the tension between the accuracy of $\hat{\theta}(X)$ and the informativeness of the auxiliary score evaluated at $\hat{\theta}(X)$. We show that the resulting pair contains, up to a smooth transformation, the unique first-order

coordinate that is invariant to the choice of pilot.

In the numerical studies, we assess exact-score and exact-posterior recovery in controlled models, while in a Smith max-stable spatial-extremes simulator we assess recovery of a scientifically meaningful spatial dependence surface.

In Models 1 and 2, NLSA reduces standardized score mean squared error by 40.6% and 32.1%, and reduces the first Wasserstein distance from the exact posterior by 24.7% and 19.2%, respectively. In the 79-site Smith experiment, it reduces the integrated squared error of the estimated extremal-dependence surface by 14.0–18.2% at the three parameter values considered.

2 Related work

Composite likelihood methods. Composite-likelihood methods construct tractable surrogate scores as linear combinations of local scores, and the literature studies the selection of their components and weights [Lindsay, 1988, Varin et al., 2011, Bai et al., 2012]. Recent theory for posterior conditioning on multiple composite scores [Li et al., 2026] shows that summary-based posterior inference can automatically yield an optimal linear combination. This work extends learning beyond the linear span of the local scores.

Architectures for learned score functions. Jiang et al. [2026] propose to learn structured statistical score functions over independent observations by imposing unbiasedness and the Bartlett information identity. Posterior-score methods instead compose conditional posterior scores over independent or Markov pieces [Geffner et al., 2023, Gloeckler et al., 2025]. Khoo et al. [2025] estimate local Fisher scores using lightweight linear parameterizations. These architectures use raw-data encoders and rely on particular likelihood factorizations. Our proposed architecture requires neither the statistical restrictions in Jiang et al. [2026] nor an exact likelihood factorization.

Score compression and learned summaries. SALLY uses the parameter score as a locally sufficient summary around a reference point [Brehmer et al., 2020]. Semi-automatic ABC approximates posterior means [Fearnhead and Prangle, 2012], while information-maximizing neural networks optimize low-dimensional summaries through an information criterion [Charnock et al., 2018]. Graph neural Bayes estimators encode spatial structure directly in a raw-data model [Sainsbury-Dale et al., 2025], and direct Fisher-score estimation provides another way to learn the score without likelihood evaluations [Khoo et al., 2025]. In contrast, the method proposed here starts from analytically available local scores and learns how they should be transformed before pooling.

Score validity and posterior validity impose different constraints. When an estimated function is subsequently treated as a score, score-matching methods impose mean-zero and curvature conditions [Jiang et al., 2026]. Neural posterior estimation has a different requirement: any deterministic summary $\mathcal{C}(X)$ defines a posterior $p\{\theta \mid \mathcal{C}(X)\}$. Unbiasedness is important for estimating-function efficiency and one-step estimation, but it is not required for a summary used in neural posterior estimation. Under standard local conditions, posterior procedures based on score summaries have Bernstein–von Mises limits whose precision is determined by the information in the summaries [Li and Fearnhead, 2018, Li et al., 2026]. We examine the finite-sample performance empirically in Section 4.

Auxiliary score-based summary statistics Auxiliary-model scores have often been used as summaries in SBI, with a pilot estimate $\hat{\theta}$ of the data-generating parameter

θ_0 used as the evaluation point [Drovandi et al., 2015, Ruli et al., 2016, Martin et al., 2019]. The bias of $\hat{\theta}$ can be influential because the local score approximation may deteriorate away from θ_0 [Brehmer et al., 2020]. It shows the bias of $\hat{\theta}$ affects the performance of SBI that the performance eventually plateaus and does not asymptote to zero error, because the local model approximation breaks down farther away from the evaluation point, and the score is no longer the sufficient statistic. Li et al. [2026] prove that $\hat{\theta}$ need only be consistent when approximate Bayesian computation (ABC) is used for posterior approximation. These approaches are not amortized, reflecting the tension between the accuracy and computational cost of obtaining $\hat{\theta}(X)$ for simulated X in SBI.

3 Methodology

3.1 Notations and setup

Let $X \sim p(\cdot | \theta)$ denote simulated data, where $\theta \in \Theta \in \mathbb{R}^p$ is a p -dimensional parameter and $p(\cdot | \theta)$ is the intractable joint density. Let $\beta \in \Theta$ be the point at which a score is evaluated, and let $\theta_0 \in \Theta$ be the data-generating parameter. Consider the prior density $\pi(\cdot)$ for θ . Denote the Fisher score evaluated at β by $g_0(X, \beta) = \nabla_\beta \log p(X | \beta)$, where ∇_β denotes the gradient with respect to β .

Let $\mathcal{J}$ be an index set for the $J = |\mathcal{J}|$ tractable local likelihood components. For component $j \in \mathcal{J}$, write X_j for the corresponding local data and define its local score at evaluation point $\beta \in \Theta \subseteq \mathbb{R}^p$ by

$$s_j(X; \beta) := \nabla_\beta \log f_j(X_j | \beta) \in \mathbb{R}^p,$$

where $f_j(\cdot | \beta)$ is the j th local density. Set $\mathcal{J}$ is partitioned into G_{grp} groups $\mathcal{U}_1, \dots, \mathcal{U}_{G_{\text{grp}}}$, chosen to reflect symmetries of the statistical model. Components having the same structural role share a group, while features that distinguish their local likelihood structure may be encoded and retained as marks. For example, in a stationary time series, pairwise scores from $(X_t, X_{t+\ell})$ are naturally grouped by lag ℓ , since their local distribution depends on ℓ but not on the absolute time t . In a spatial model, pairwise scores may be grouped by distance class, while the direction of the spatial lag $h_{ij} = x_j - x_i$ is retained as a mark, together with the local score to form the local component when anisotropy makes direction statistically relevant. Hence within each such group, the ordering of the local components is regarded as arbitrary. Accordingly, all feature representations of the local components are required to be invariant under independent within groups permutations. If component j is associated with a mark, denoted by a_j , write $z_j(X; \beta) := \{s_j(X; \beta), a_j\}$; otherwise $z_j = s_j$.

We also let $b = 1, \dots, K$ index outer blocks. We take $K > 1$ only when the Fisher score admits an additive decomposition $g_0(X, \beta) = \sum_{b=1}^K g_b(X, \beta)$; otherwise $K = 1$ and $g_1 = g_0$. For group g in block b , define

$$\mathcal{U}_{g,b} := \{j \in \mathcal{U}_g : X_j \text{ lies in block } b\}, U_{g,b}(X; \beta) := \{z_j(X; \beta) : j \in \mathcal{U}_{g,b}\},$$

and write $U_b(X; \beta) := (U_{1,b}, \dots, U_{G_{\text{grp}},b})(X; \beta)$ for the grouped local-component collection in block b , and $U(X; \beta) := (U_1(X; \beta), \dots, U_K(X; \beta))$ for the complete collection. Examples of these groupings are given in Appendix A and J.

Define

$$\mathbb{L} = \overline{\text{span}}\{s_{j,k}(X; \theta_0) : j \in \mathcal{J}, k = 1, \dots, p\} \subset L_0^2\{\sigma(U)\}$$

as the closed scalar linear span of all local-score coordinates, where $s_{j,k}$ denotes coordinate k of s_j and $L_0^2\{\sigma(U)\}$ contains the mean-zero square-integrable functions measurable with respect to $\sigma(U)$.

By the estimating function theory, for an unbiased estimating function $f(X, \theta) \in \mathbb{R}^{d_f}$ measurable with respect to $\sigma(U)$ and having positive-definite covariance $V_f(\theta) = \text{Var}_\theta(f)$, define its sensitivity and Godambe information by $H_f(\theta) = \mathbb{E}_\theta\{f(X, \theta) g_0(X, \theta)^\top\}$ and $G(f) = H_f^\top V_f^{-1} H_f$.

3.2 Structured score architecture

Under the expected score-matching loss, the optimal score approximation square-integrable and measurable with respect to $U(X; \beta)$ is

$$m_U\{U(X; \beta), \beta\} := \mathbb{E}_X\{g_0(X, \beta) \mid U(X; \beta), \beta\} = \underset{S \in L^2(\sigma(U))}{\text{argmin}} \mathbb{E}_{\beta, X} \|S\{U(X; \beta), \beta\} - g_0(X, \beta)\|^2, \quad (1)$$

where $\mathbb{E}_{\beta, X}$ denotes the expectation under $\beta \sim \pi(\cdot), X \sim p(\cdot \mid \beta)$. The equation above holds by the L^2 projection argument and the fact that under the law of total expectation with conditioning on $(U(X; \beta), \beta)$, the cross term vanishes since $\mathbb{E}_X\{m_U(U, \beta) - g_0(X, \beta) \mid U(X; \beta), \beta\} = 0$. Moreover, let $q_\beta(u; \theta)$ be probability density of $U(X; \beta)$ implied by $X \sim p(\cdot \mid \theta)$.

By Fisher's identity with X as the latent variable, under the usual domination and differentiation-under-the-integral conditions, $m_U(U, \beta)$ is also equal to $\nabla_\theta \log q_\beta(U \mid \theta)|_{\theta=\beta}$, i.e. the Fisher score for the distribution of U . By this fact, $I_U(\theta) := \text{Var}_\theta\{m_U(U, \theta)\}$ is the implied Fisher information of $U(X; \theta)$, and for any measurable retained representation $T(X; \theta) = T(U)$, $I_T(\theta)$ is its implied Fisher information. Under these notations the following proposition explains why $m_U(U, \beta)$ is chosen as the population target of our proposed architecture.

Proposition 1 (Information hierarchy for representations). Assume $\mathbb{E}_\theta \|g_0(X, \theta)\|^2 < \infty$. For every measurable representation $T = T(U)$, each unbiased square-integrable estimating function $f = f(T)$ with positive-definite covariance satisfies

$$G(f) \preceq I_T \preceq I_U \preceq I(\theta),$$

where $\preceq$ is the Loewner order and $I(\theta) := \text{Var}_\theta\{g_0(X, \theta)\}$, the Fisher information. The first equality holds with the choice $f = m_T$ whenever I_T is nonsingular. The second equality holds when $m_U(U)$ is measurable with respect to $\sigma(T)$.

Proof. The first inequality holds by the fact that $H_f^\top V_f^{-1} f$ is the best linear projection of $m_T(T)$ onto the linear span of the components of f [Heyde, 1997, Chapter 2], and $G(f) = \text{Var}(H_f^\top V_f^{-1} f)$. Since $T = T(U)$, $m_T(T) = \mathbb{E}_{\theta_0}\{m_0(U) \mid T\}$, and hence by the law of total covariance, the second inequality holds by

$$I_U - I_T = \mathbb{E}_{\theta_0} \left[(m_U - m_T)(m_U - m_T)^\top \right] \succeq 0.$$

Likewise, the third holds by

$$I(\theta) - I_U = \mathbb{E}_{\theta_0} \left[(g_0 - m_U)(g_0 - m_U)^\top \right] \succeq 0.$$

The equality statements follow from vanishing projection residuals. $\square$

Proposition 1 says that $m_U(U, \theta)$ has the maximal information among all $T(U)$ and all linear combinations of any unbiased estimating function $f(T)$. The classical composite likelihood theory targets to maximize $G(f(U))$, the information of the best linear projection of $g_0(X; \theta)$ onto $\mathbb{L}$. Therefore we have the projection decomposition: $I(\theta_0) - I_U = \mathbb{E}_{\theta_0} \{\text{Var}(g_0(X, \theta_0) | U)\}$, is the information in the full data that is not available from the local scores; $I_U - G_{\text{lin}}^* = \text{Var}_{\theta_0} \{m_0(U) - \Pi_{\mathbb{L}} g_0(X, \theta_0)\}$, the nonlinear-versus-linear information gap; and $G_{\text{lin}}^* = \text{Var}_{\theta_0} \{\Pi_{\mathbb{L}} g_0(X, \theta_0)\}$, where $\Pi_{\mathbb{L}}$ denote scalar L^2 projection onto $\mathbb{L}$, applied coordinatewise to vector-valued functions, the Godambe information of the optimal linear combination of $U(X; \theta)$.

For block size $K > 1$, the additive decomposition of the Fisher score implies the population target $m_U(U, \beta)$ also admits the additive decomposition,

$$m_U(U, \beta) = \sum_{b=1}^K m_b(U_b, \beta), \quad m_b(U_b, \beta) := \mathbb{E}_X \{g_b(X, \beta) | U_b(X, \beta), \beta\}.$$

For each group b , because $U_b(X; \beta)$ is defined as an unordered grouped collection, any measurable function of U_b , in particular the conditional block score $m_b(U_b, \beta)$ is invariant to the corresponding independent within-group permutations.

This invariance suggests the following transform-sum-readout architecture for the surrogate score:

$$S_\omega(X, \beta) = R_\psi(R_1, \dots, R_G, \beta) \in \mathbb{R}^p, \quad R_g(X, \beta) = \sum_{j \in \mathcal{U}_g} \phi_{\eta, g}\{s_j(X; \beta), \beta\}, \quad (2)$$

where $\phi_{\eta, g}$ is a nonlinear map shared in group g with parameters η , R_ψ is a readout with parameters ψ , and $\omega = (\psi, \eta)$. Summation makes the representation invariant to relabelling within each group, as in grouped Deep Sets [Zaheer et al., 2017, Wagstaff et al., 2022]. $\phi_{\eta, g}(s, \beta)$ being a linear function gives the classical composite scores method, since a flexible outer readout R_ψ does not change the first-order information. This holds intuitively by the Delta method under the standard assumptions that the centered and scaled $(R_1, \dots, R_G)$ converge in distribution.

If $\phi_{\eta, g}(s, \beta) = A_g(\beta)s$ is linear, then $R_g = A_g(\beta)C_g$ where $C_g = \sum_{j \in \mathcal{U}_g} s_j(X; \beta)$, so the outer readout R_ψ sees only the usual group-wise linear composite-score pools. Under their standard Gaussian local limit, the Fisher score of $C = (C_1, \dots, C_G)$ is itself linear in C . Thus a nonlinear readout applied only after this pooling cannot improve first-order information beyond the optimal weighted composite score. Nonlinear $\phi_{\eta, g}$ matters because it acts before summation, and can therefore preserve features of the individual local scores that linear pooling would discard. see Appendix C for a formal statement.

For the class of $S_\omega(X, \beta)$ to achieve universal approximation, $\phi_{\eta, g}$ should be expressive enough. In experiments for group g we choose $\phi_{\eta, g}(s, \beta) = (s, m_{\eta, g}(s, \beta))$ where $m_{\eta, g}(s, \beta)$ is a multilayer perceptron (MLP), with which $S_\omega(X, \beta)$ is our proposed *Nonlinear Local-Score Aggregation (NLSA)* architecture. The training of $\phi_{\eta, g}(s, \beta)$ is initialized from $\phi_{\eta, g}(s, \beta) = s$, called *Identity Local-Score Aggregation (ILSA)*, which applies the nonlinear

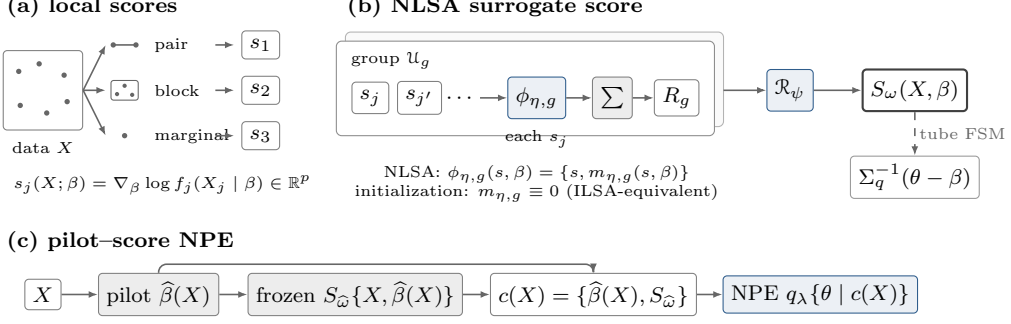


Figure 1: NLSA Pilot-Score NPE. Shaded blue components are learned. Within an additive block structure, the transform-sum-readout operation is applied per block and the resulting score contributions are summed as in Eq. (2).

outer readout R_{ψ} on multiple composite score functions. In practice, a small SiLU MLP for $m_{\eta, g}$ performs well. As the examples in Appendix A show, the optimal nonlinear map shared within a local score group can be very simple.

3.3 Estimation by tube Fisher-score matching

For a single dataset $X_{\text{obs}} \sim p(\cdot | \theta_0)$, we are interested in the variation of the Fisher score, $\text{Var}\{g_0(X_{\text{obs}}, \theta_0)\}$, where $g_0(X, \theta) = \nabla_{\theta} \log p(X | \theta)$, which gives the Fisher information. For the pointwise optimization, one criterion function to be minimized is the score-matching L2 loss

$$L_{\theta_0}(\omega) = E_{X \sim p(\cdot | \theta_0)} \|S_{\omega}(X) - g_0(X, \theta_0)\|^2,$$

and under a flexible enough class, the optimizer is $S_{\omega^*}(X) = g_0(X, \theta_0)$ so its variation gives the Fisher information. So if X_{obs} contains enough independent copies, the empirical version of $L_{\theta_0}(\omega)$ is available, and the optimizer $\tilde{\omega}$ can be obtained. Its direct optimization gives that, towards $g_0(X, \theta_0)$, $S_{\tilde{\omega}}(X)$ is more accurate in higher probability region of X_{obs} than the lower probability region.

If the function class is restricted under the representation $S_{\omega}\{U(X)\}$, the optimizer is $S_{\omega^*}(u) = E_{X \sim p(\cdot | \theta_0)}\{g_0(X, \theta_0) | U(X) = u\}$. Then the law of total covariance gives the loss of information by $\text{Var}_X\{g_0(X, \theta_0)\} - \text{Var}_X\{S_{\omega^*}(U)\} = E_X[\text{Var}_X\{g_0(X, \theta_0) | U(X)\}]$ which is the cost of using a more structured and compressed feature $U(X)$ of X .

The above optimization builds the local geometry of the likelihood at θ_0 into ω^* , so is not amortized, i.e. each dataset with different data-generating parameter requires to re-train ω^* . An amortized inference means that a pre-trained function can be used on any given dataset X to approximate the Fisher score $g_0(X, \theta_X)$ where θ_X is the unknown data-generating parameter. For the amortized inference, the function class should be $S_{\omega}(X, \theta_X)$, also depending on θ_X , since for the same X but different θ_X , the optimizer should fit different geometry. We can minimize the following score-matching loss,

$$L(\omega) = E_{\beta \sim \pi_{\beta}(\cdot), X \sim p(\cdot | \beta)} \|S_{\omega}(X, \beta) - g_0(X, \beta)\|^2. \quad (3)$$

where π_{β} is a distribution on evaluation points. Similarly, for a flexible enough class, the optimizer is $g_0(X, \beta)$, and for the class restricted by $U(X)$, it is

$$S_{\omega^*}(u, \beta) = m_U(u, \beta) = E_{X \sim p(\cdot | \beta)}\{g_0(X, \beta) | U(X) = u, \beta\},$$

where m_U is defined in (1). Note that by using simulations, we don't need that X contains independent copies. The empirical optimizer $\tilde{\omega}$ gives more accurate estimate $S_{\tilde{\omega}}(X, \beta)$ in the higher density area of (β, X) towards $g_0(X, \beta)$. We call the area in $(\theta, \beta) \in \Theta \times \Theta$ the 'diagonal' region where the generating parameter θ of X is the same as the score evaluating parameter β .

The training of S_{ω} can be done through the forward score-matching (FSM) regression extending the point-wise matching at a fixed β of Khoo et al. [2025] to a 'tube' area for amortization, i.e. the generating parameter θ is different from the evaluating parameter β . Let $q_{\text{tube}}(\theta | \beta) = \mathcal{N}_p(\theta; \beta, \Sigma_q)$, where $\Sigma_q \in \mathbb{R}^{p \times p}$ is positive definite. We draw $\beta \sim \pi_{\beta}$, $\theta \sim q_{\text{tube}}(\cdot | \beta)$, $X \sim p(\cdot | \theta)$, and minimize

$$\mathcal{L}_{\text{FSM}}(\omega) = \mathbb{E} [\|S_{\omega}(X, \beta) - \nabla_{\beta} \log q_{\text{tube}}(\theta | \beta)\|^2], \quad (4)$$

$$\text{equivalent to } \mathbb{E}_{X \sim p_{\text{tube}}(\cdot | \beta), \beta \sim \pi(\cdot)} [\|S_{\omega}(X, \beta) - \nabla_{\beta} \log p_{\text{tube}}(X | \beta)\|^2],$$

where $\nabla_{\beta} \log q_{\text{tube}}(\theta | \beta) = \Sigma_q^{-1}(\theta - \beta)$. For fixed β , this is the local FSM loss of Khoo et al. [2025]. Averaging over β learns the surface of the target conditional Fisher score regarding the generating parameter smoothed by the Gaussian kernel. Specifically, the population target of (4) is

$$s_{\Sigma_q}(X, \beta) = \mathbb{E}_{\theta \sim p_{\text{tube}}(\cdot | X, \beta)} \{g_0(X, \theta) | X, \beta\} = \nabla \log p_{\text{tube}}(X | \beta),$$

$$\text{where } p_{\text{tube}}(x | \beta) = \int p(x | \theta) q_{\text{tube}}(\theta | \beta) d\theta \text{ and } p_{\text{tube}}(\theta | x, \beta) = \frac{p(x | \theta) q_{\text{tube}}(\theta | \beta)}{p_{\text{tube}}(x | \beta)}.$$

In Khoo et al, the local surrogate score is retrained at every iteration of the likelihood maximisation for a given dataset. Here the loss (4) is used for amortized inference: the surrogate score is trained once for $\beta \sim \pi(\cdot)$ and be deployed to observed dataset with any generating parameter θ .

Under the class restricted by $U(X)$, it is

$$m_{\Sigma_q}(u, \beta) := \mathbb{E}_{X \sim p_{\text{tube}}(\cdot | \beta)} \{\nabla_{\beta} \log p_{\text{tube}}(X | \beta) | U(X) = u, \beta\},$$

and as $\Sigma_q \rightarrow 0$, it approaches $m_U(u, \beta)$. Further explanation of the amortized local FSM is given in Appendix D.

Say the trained parameter $\tilde{\omega}$ is obtained. At deployment for a given dataset X_{obs} , ideally $S_{\tilde{\omega}}(X_{\text{obs}}, \theta)$ can be evaluated which however depends on the unknown generating parameter θ of X_{obs} . The surrogate score will be evaluated at some random evaluation point $\hat{\beta}_{\theta}$ deviating from θ , i.e. $S_{\tilde{\omega}}(X_{\text{obs}}, \hat{\beta}_{\theta})$. Let $\Sigma_e = E_{\theta, \hat{\beta}}\{(\hat{\beta}_{\theta} - \theta)(\hat{\beta}_{\theta} - \theta)^T\}$, the variation of the error of $\hat{\beta}_{\theta}$, $\Sigma_q = c\Sigma_e$, where c is a scaling constant, and $s_c(X, \beta) = s_{\Sigma_q}(X, \beta)$. The error approximating the Fisher score evaluated at β can be decomposed as

$$S_{\tilde{\omega}}(X, \beta) - g_0(X, \beta) = \underbrace{S_{\tilde{\omega}}(X, \beta) - s_c(X, \beta)}_{\text{learned-surface error}} + \underbrace{s_c(X, \beta) - g_0(X, \beta)}_{\text{smoothing error}}.$$

For the smoothing error term, consider the simplified kernel variance $\Sigma_q = cI_p$. Khoo et al. [2025] gives the following upper bound:

$$E_{X \sim p(\cdot | \theta)} \|s_c(X, \beta) - g_0(X, \beta)\| \leq L\sqrt{pc}E_{x \sim p(\cdot | \theta)}, \left[\frac{p(x | \theta)}{p(x | \beta)} \right]. \text{ (to be updated to the random } \hat{\beta} \text{ setting)}$$

It shows that the smoothing error decreases as c decreases and goes to zero as $c \rightarrow 0$. Meanwhile on the off-diagonal where $\theta \neq \beta$, the error additionally depends on the expected likelihood ratio $p(x|\theta)/p(x|\beta)$ which increases as the overlap between the two densities gets smaller, roughly speaking, as β gets further away from θ .

Learned-surface error. The tube covariance controls how accuracy under the tube-training distribution transfers to observations generated at θ when the score is evaluated at a displaced parameter β . Let $\{p(\cdot | \vartheta) : \vartheta \in \mathbb{R}^p\}$ be densities with respect to a common dominating measure μ . For a fixed positive-definite pilot-error scale matrix $M_e \in \mathbb{R}^{p \times p}$, define

$$q_c(\vartheta | \beta) = \mathcal{N}_p(\vartheta; \beta, c^2 M_e), \quad p_c(x | \beta) = \int p(x | \vartheta) q_c(\vartheta | \beta) d\vartheta,$$

where $c > 0$ is a dimensionless standard-deviation multiplier, so that $\Sigma_q = c^2 M_e$. Write $s_c(x, \beta) = \nabla_\beta \log p_c(x | \beta)$, assuming the relevant derivatives and expectations exist.

Condition on the fitted surrogate $S_{\tilde{\omega}}$, and let $B = \hat{\beta}_\theta \sim \Pi_\theta$ be independent of $X \sim p(\cdot | \theta)$. Define

$$e_c(x, \beta) = S_{\tilde{\omega}}(x, \beta) - s_c(x, \beta), \quad \varepsilon_c(\beta) = \mathbb{E}_{X \sim p_c(\cdot | \beta)} \|e_c(X, \beta)\|,$$

and the coverage factor

$$C_c(\theta, \beta) = \text{ess sup}_x \frac{p(x | \theta)}{p_c(x | \beta)}.$$

The essential supremum is taken with respect to μ , with the ratio interpreted as infinite wherever the numerator is positive and the denominator vanishes. Under domination,

$$\begin{aligned} \mathbb{E}_B \mathbb{E}_{X \sim P_\theta} \|e_c(X, B)\| &= \mathbb{E}_B \mathbb{E}_{X \sim p_c(\cdot | B)} \left[\frac{p(X | \theta)}{p_c(X | B)} \|e_c(X, B)\| \right] \\ &\leq R_c := \mathbb{E}_B [C_c(\theta, B) \varepsilon_c(B)]. \end{aligned} \tag{5}$$

Here $C_c(\theta, \beta)$ measures the worst-case inflation in transferring an error bound from the tube-training distribution to the deployment distribution P_θ , while $\varepsilon_c(\beta)$ is the population training error of the surrogate score under the tube-FSM. Different from the smoothing error where smaller c is better, as $c \rightarrow 0$, $C_c(\theta, \beta) \rightarrow \sup_x p(x|\theta)/p(x|\beta)$ which is usually infinity for many continuous model when $\theta \neq \beta$. The Gaussian location family is a simple example. The proposition below provides sufficient conditions for the boundedness of the coverage factor.

Proposition 3.1 (Local-curvature coverage bound). *Fix $\theta \in \mathbb{R}^p$ and write $\ell_x(\theta) = \log p(x | \theta)$. Assume there exist constants $L_\theta \geq 0$ and $\rho > 0$ such that ℓ_x is finite and twice continuously differentiable on a neighbourhood of*

$$\mathcal{E}_{\theta, \rho} = \{\vartheta : \|M_e^{-1/2}(\vartheta - \theta)\| \leq \rho\},$$

and

$$\nabla_\vartheta^2 \ell_x(\vartheta) \succeq -L_\theta M_e^{-1}, \quad \vartheta \in \mathcal{E}_{\theta, \rho}.$$

Let $F_{\chi_p^2}$ denote the distribution function of a chi-squared random variable with p degrees

of freedom, and define

$$\tau_{c,\rho} = 1 - F_{\chi_p^2} \left(\frac{\rho^2(1 + L_\theta c^2)}{c^2} \right), \quad H_{c,\rho} = \frac{(1 + L_\theta c^2)^{p/2}}{1 - \tau_{c,\rho}}.$$

Then, for every fixed $\beta \in \mathbb{R}^p$ and $c > 0$,

$$C_c(\theta, \beta) \leq H_{c,\rho} \exp \left\{ \frac{\|M_e^{-1/2}(\theta - \beta)\|^2}{2c^2} \right\} < \infty.$$

Proof. Fix β and set $d = M_e^{-1/2}(\theta - \beta)$ and $v_\theta(x) = M_e^{1/2} \nabla_\theta \ell_x(\theta)$. Under the proposal, $U = M_e^{-1/2}(\vartheta - \theta) \sim \mathcal{N}_p(-d, c^2 I_p)$. Taylor's theorem and the curvature assumption imply

$$\ell_x(\theta + M_e^{1/2}u) - \ell_x(\theta) \geq v_\theta(x)^\top u - \frac{L_\theta}{2} \|u\|^2, \quad \|u\| \leq \rho.$$

Let $\varphi_{m,V}$ denote the density of $\mathcal{N}_p(m, V)$. Restricting the nonnegative mixture integral to $\|u\| \leq \rho$, then pairing u and $-u$ and applying the arithmetic–geometric mean inequality, gives

$$\begin{aligned} \frac{p_c(x \mid \beta)}{p(x \mid \theta)} &\geq \int_{\|u\| \leq \rho} e^{v_\theta(x)^\top u - L_\theta \|u\|^2/2} \varphi_{-d, c^2 I_p}(u) du \\ &\geq \int_{\|u\| \leq \rho} e^{-L_\theta \|u\|^2/2} \sqrt{\varphi_{-d, c^2 I_p}(u) \varphi_{-d, c^2 I_p}(-u)} du \\ &= e^{-\|d\|^2/(2c^2)} (2\pi c^2)^{-p/2} \int_{\|u\| \leq \rho} e^{-(L_\theta + c^{-2})\|u\|^2/2} du \\ &= e^{-\|d\|^2/(2c^2)} (1 + L_\theta c^2)^{-p/2} F_{\chi_p^2} \left(\frac{\rho^2(1 + L_\theta c^2)}{c^2} \right). \end{aligned}$$

The final probability is strictly positive for $c, \rho > 0$. Taking reciprocals and the essential supremum over x proves the result. $\square$

The sufficient conditions for the deployment error to be finite are given in the following corollary.

Corollary 2 (Average transfer for an independent pilot). Under Proposition 3.1, let $B \sim \Pi_\theta$ be independent of $X \sim P_\theta$, and define

$$D = M_e^{-1/2}(\theta - B), \quad \mathcal{T}_c^2 = \mathbb{E}_B \mathbb{E}_{X \sim p_c(\cdot \mid B)} \|e_c(X, B)\|^2.$$

For any fixed $c > 0$ satisfying

$$\mathcal{T}_c^2 < \infty, \quad \mathbb{E}_B e^{\|D\|^2/c^2} < \infty, \quad (6)$$

the coverage-weighted learning error in (5) satisfies

$$\mathbb{E}_B \mathbb{E}_{X \sim P_\theta} \|e_c(X, B)\| \leq R_c \leq H_{c,\rho} \left(\mathbb{E}_B e^{\|D\|^2/c^2} \right)^{1/2} \mathcal{T}_c < \infty.$$

Proof. Jensen's inequality gives $\mathbb{E}_B \varepsilon_c(B)^2 \leq \mathcal{T}_c^2$. Proposition 3.1 and Cauchy–Schwarz

therefore yield

$$R_c \leq H_{c,\rho} \mathbb{E}_B \left[e^{\|D\|^2/(2c^2)} \varepsilon_c(B) \right] \leq H_{c,\rho} \left(\mathbb{E}_B e^{\|D\|^2/c^2} \right)^{1/2} \mathcal{T}_c.$$

The deployment-error bound follows from (5). $\square$

Condition (6) assumes the finite learning error averaged over B and that the tube bandwidth is sufficiently wide so that the integrand is integrable over the tail regions of D . One sufficient condition for the latter is that the tail of the centered standardized pilot error is sub-Gaussian with scale $\kappa > 0$, meaning the upper bounding of the moment generating function of D , $\mathbb{E} e^{t^\top D} \leq e^{\kappa^2 \|t\|^2/2}$ for every $t \in \mathbb{R}^p$. For any scalar c satisfying $\sqrt{2}\kappa < c < \infty$, this implies $\mathbb{E}_B e^{\|D\|^2/c^2} \leq (1 - 2\kappa^2/c^2)^{-p/2} < \infty$ (add citation Hsu, D., Kakade, S. M., and Zhang, T. (2012). A tail inequality for quadratic forms of subgaussian random vectors). As an example, for an exactly Gaussian standardized B , $D \sim \mathcal{N}_p(0, I_p)$, then $\kappa = 1$, giving the sufficient condition $c > \sqrt{2}$.

In practice, a pilot estimate $\hat{\theta}(X)$ of the generating parameter θ is constructed and the surrogate score is deployed as $S_w(X, \hat{\theta}(X))$. The tube kernel bandwidth Σ_q is initialized to be $2\Sigma_e$ estimated by pilot simulations, motivated by assuming asymptotic Normality of $\hat{\theta}(X)$. Then the bandwidth is further tuned to balance the bias-variance trade-off of the training and smoothing error following the procedure suggested by Khoo et al. [2025].

Remark 3.1. Direct parameter score matching requires differentiating $S_{\psi,\eta}(X, \theta)$ with respect to θ . The resulting Jacobian differentiates every local score and therefore requires second derivatives of the local log likelihoods. Tube regression instead uses the known proposal score and avoids these derivatives.

3.4 Joint use of the pilot and the estimated score

The following local expansion explains why off-diagonal training is relevant when a root- n consistent pilot is used. Let g_n denote the score based on a sample of effective size n , and let $\mathcal{I}_1(\theta_0)$ denote the Fisher information per effective observation. For $\beta = \theta_0 + e_\beta$, where $e_\beta \in \mathbb{R}^p$ is a local displacement,

$$g_n(X, \theta_0 + e_\beta) \approx g_n(X, \theta_0) - n\mathcal{I}_1(\theta_0)e_\beta.$$

The stochastic term is $O_p(\sqrt{n})$, whereas the displacement term is $O_p(n\|e_\beta\|)$. For a root- n consistent pilot, $e_\beta = O_p(n^{-1/2})$, and hence the two terms have the same order. The one-step coordinate introduced in the next subsection corrects this displacement to first order.

Let $\hat{\beta}(X)$ be a pilot estimate computed from the data. We use the following summary for posterior inference:

$$Z_{\text{pair}}(X) = \{\hat{\beta}(X), S_{\psi,\eta}(X, \hat{\beta}(X))\} \in \mathbb{R}^{2p}. \quad (7)$$

Define the centered estimating function

$$b(\beta) := \mathbb{E}_\beta \{S_{\psi,\eta}(X, \beta)\}, \quad \Psi(X, \beta) := S_{\psi,\eta}(X, \beta) - b(\beta).$$

Because $b(\beta)$ is a deterministic function of β , replacing $S_{\psi,\eta}$ by Ψ in Z_{pair} is a one-to-one

re-expression of the same summary. Define

$$\mu(\theta, \beta) := \mathbb{E}_\theta\{\Psi(X, \beta)\}, \quad H(\beta) := \partial_\theta \mu(\theta, \beta)|_{\theta=\beta},$$

where $H(\beta) \in \mathbb{R}^{p \times p}$ is the sensitivity matrix, and define the variability and Godambe information matrices

$$V(\beta) := \text{Var}_\beta\{\Psi(X, \beta)\}, \quad G(\beta) := H(\beta)^\top V(\beta)^{-1} H(\beta),$$

when $V(\beta)$ is nonsingular. By construction, $\mu(\theta, \theta) = 0$ for every θ , so centering gives Fisher consistency on the diagonal. Let $H_0 = H(\theta_0)$ and assume that H_0 is nonsingular and that the usual smooth local expansion is valid. Differentiating the diagonal identity gives

$$\partial_\beta \mu(\theta, \beta)|_{\theta=\beta=\theta_0} = -H_0.$$

Thus the score value depends to first order on the point at which it is evaluated. Evaluation at θ_0 avoids displacement due to the pilot, whereas $\Psi\{X, \widehat{\beta}(X)\}$ contains this displacement. At the other extreme, if $\widetilde{\beta}(X)$ is a root satisfying $\Psi\{X, \widetilde{\beta}(X)\} = 0$, the score value is identically zero and contains no information unless the root itself is retained.

The next two propositions show why the evaluation point should be retained. We use the first-order local translation model

$$d\Psi = H_0 d\theta - H_0 d\beta,$$

where the vectors $d\theta, d\beta \in \mathbb{R}^p$ are infinitesimal perturbations of the generating parameter and evaluation point, respectively. This relation is the local form of the diagonal identity above.

For a differentiable statistic F , we say that it is *first-order informative* in a generating direction $d\theta$ if its differential is nonzero in that direction. It is first-order uninformative if its differential is zero for every $d\theta$.

Proposition 3 (Score-only pilot-invariance impossibility). Let H_0 be nonsingular and let d_F be a positive integer and $F : \mathbb{R}^p \rightarrow \mathbb{R}^{d_F}$ be differentiable, with Jacobian F_Ψ at the local score value. Under the local translation model, if $F(\Psi)$ is first-order invariant to perturbations $d\beta$ of the evaluation point, then it has zero first-order response to perturbations $d\theta$ of the generating parameter. Hence a differentiable statistic of the score value alone that is first-order pilot-invariant is first-order uninformative in every generating-parameter direction.

Proposition 4 (Pilot-invariant pair characterization). Let d_F be a positive integer and $F : \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}^{d_F}$ be continuously differentiable, and write F_β and F_Ψ for its two Jacobians. Under the same local translation model, first-order invariance to $d\beta$ is equivalent to

$$F_\beta = F_\Psi H_0.$$

Locally, every solution has the form

$$F(\beta, \Psi) = \Xi\{\beta + H_0^{-1}\Psi\}$$

for a differentiable function $\Xi : \mathbb{R}^p \rightarrow \mathbb{R}^{d_F}$. Thus $\beta + H_0^{-1}\Psi$ is, up to a smooth transformation, the first-order pilot-invariant coordinate of the pair.

Proposition 4 identifies the one-step coordinate

$$\widehat{\beta}_1 = \widehat{\beta} + H_0^{-1}\Psi(X, \widehat{\beta})$$

as the canonical first-order pilot-invariant reduction of the pair. Under the usual conditions for one-step estimation, including $\|\widehat{\beta} - \theta_0\| = o_p(n^{-1/4})$ and a sufficiently smooth sensitivity matrix,

$$\widetilde{\beta}(X) = \widehat{\beta}_1 + o_p(n^{-1/2}).$$

More Newton steps may be used when the pilot is less accurate. Let $\sigma(Y)$ denote the sigma-field generated by a random quantity Y . Then

$$\sigma(\widehat{\beta}_1) \subseteq \sigma\{\widehat{\beta}, \Psi(X, \widehat{\beta})\},$$

so the pair contains the one-step summary and may also retain higher-order finite-sample information.

For the exact Fisher score, H_0 is the Fisher information matrix and can be estimated from the score covariance. For a general centered approximation, $H(\beta)$ can be estimated by finite differences of $\mathbb{E}_\theta\{\Psi(X, \beta)\}$ around $\theta = \beta$. Our neural posterior estimation experiments condition on the full pair and therefore do not require an estimate or inverse of H .

The two stages are estimated separately because the posterior objective identifies only the final summary supplied to the posterior estimator. It does not separately identify the local transformation inside the many-to-one pooling operation. We therefore estimate and fix this transformation in Stage 1 before fitting the posterior estimator in Stage 2.

Algorithm 1 describes both stages.

4 Numerical studies

Table 1: Summary of the empirical comparisons. Rows marked as outstanding require additional experiments.

Claim	Evidence	Status
Local nonlinear features	ILSA versus NLSA in Models 1–2	Current
Smith model	Extremal-dependence recovery	Current
Modern raw-data SBI base-lines	Flow NPE and neural posterior-score budget curves	Outstanding
Pooling boundary and capacity	Blockwise versus global pooling; capacity-scaled ILSA	Outstanding
Gain beyond the linear span	Multi-parameter exact-reference model	Outstanding
Tube and pair design choices	Tube-width and pair-versus-one-step comparisons	Outstanding

OUTSTANDING WORK. Compare local-score methods with modern flow and posterior-score baselines at matched simulation budgets.

OUTSTANDING WORK. Test the pooling boundary and a capacity-scaled ILSA readout.

OUTSTANDING WORK. Test a multi-parameter model with an exact score to isolate gains beyond the linear span.

OUTSTANDING WORK. Measure tube-width sensitivity and compare the pilot-plus-score pair with one-step and oracle-evaluation-point summaries. Repeat Smith Stage 1 across training seeds.

We use three examples to investigate whether transforming local likelihood scores before pooling improves estimation of the full-data score and the resulting posterior inference. In Models 1 and 2, both the full score and the exact posterior are available for evaluation but are not used in training or model selection. These examples therefore allow direct assessment of the two stages of the method. The third example is a Smith max-stable model for spatial extremes. Its full likelihood is unavailable, but its exact bivariate likelihoods can be evaluated. We use this example to assess estimation of the spatial extremal-dependence surface.

4.1 Comparison design

Stage-1 score comparisons. We compare three methods for estimating the score as a function of the data and evaluation point. **Raw-data FSM** predicts the forward score-matching (FSM) target directly from the observations. The other two methods use the same tractable local likelihood scores. **ILSA** pools these scores without a learned local transformation, whereas **NLSA** appends a learned nonlinear feature to each score before using the same groups, pooling operation, and readout. Thus the comparison between ILSA and NLSA isolates the effect of the transformation before pooling. For Models 1 and 2, we compare each estimated score with the exact full score on held-out data.

Stage-2 posterior comparisons. In Stage 2, Raw-data FSM, ILSA, and NLSA use the same family of neural posterior estimators, conditioned on the data-derived pilot and the corresponding fixed Stage-1 score function. We also include **Raw-data NPE**, where NPE denotes neural posterior estimation, which estimates the posterior directly from the observations, and **Pilot-only NPE**, which conditions only on the pilot. Because these two methods change either the representation or the two-stage construction, they provide broader comparisons rather than a direct comparison of NLSA with ILSA. For the Smith model, the **all-pair MPLE**, or maximum pairwise-likelihood estimator, is included as a classical point-estimation benchmark. The neural baselines are defined in Appendix G, and the Smith MPLE is described in Appendix J.

4.2 Controlled examples

Both models have a scalar mixing probability π as the parameter of interest, with $u = \text{logit}(\pi)$ used as an unconstrained working parameter. Model 1 consists of 20 independent mean-shift mixture blocks, each of size 20, and uses observation-level scores as the local contributions. Its simple form makes it possible to see analytically why a nonlinear transformation before pooling is useful. Model 2 consists of 40 blocks of size 40 and introduces dependence within each block through a common latent factor. Marginal and pairwise scores are pooled separately, so this model assesses the same comparison in the presence of dependence and two types of local likelihood. The full likelihood is tractable in both examples, which gives exact full scores and posterior distributions for evaluation. Neither is used for training or checkpoint selection. Full details of the models and pilots are given in Appendix I.

Evaluation. For each held-out dataset and evaluation point, we compare the estimated full score with its exact value. We report the mean squared error (MSE) divided by the variance of the exact score at the same generating value. This standardized MSE is

comparable across different values of π . The parameter-specific results in the appendix report the unstandardized MSE. Their 95% t intervals are calculated from five independent repetitions of Stage-1 training and reflect variation between training runs.

For Stage 2, we compare each estimated posterior with the exact posterior for the same test dataset. The one-dimensional first Wasserstein distance, denoted by W_1 , measures the discrepancy between the two distributions, while the posterior-mean MSE compares their means. We assess posterior uncertainty using the absolute error in posterior standard deviation (SD) and the empirical coverage of nominal 90% credible intervals. The metrics are defined in Appendix L.

Exact-score recovery. Table 2 shows that NLSA has the smallest standardized score MSE in both models. Compared with ILSA, the reduction is 40.6% in Model 1 and 32.1% in Model 2. After averaging over the five Stage-1 training repetitions, its error is smaller at every value of π considered. Raw-data FSM has standardized score MSEs of 0.4615 and 1.0037 in Models 1 and 2, compared with 0.0770 and 0.1575 for ILSA. Thus, in these examples, using the tractable local likelihoods is substantially more accurate than estimating the score directly from the raw data. Figure 3 in the appendix gives the results separately for each value of π . The improvement from NLSA is already present in Stage 1, before the neural posterior estimator is fitted.

Table 2: Exact-score recovery. Entries are mean $\pm$ sample SD of standardized score MSE across five Stage-1 training repetitions; lower is better. Bold and underline mark the best and second-best methods within each model.

Method	Model 1	Model 2
Raw-data FSM	0.4615 $\pm$ 0.0144	1.0037 $\pm$ 0.0032
ILSA	<u>0.0770 $\pm$ 0.0034</u>	<u>0.1575 $\pm$ 0.0087</u>
NLSA	0.0457 $\pm$ 0.0054	0.1070 $\pm$ 0.0032

Posterior accuracy. Table 3 shows that NLSA gives the most accurate posterior approximation according to both metrics in each model. Compared with ILSA, it reduces W_1 by 24.7% in Model 1 and 19.2% in Model 2, and reduces the posterior-mean MSE by 46.8% and 34.9%, respectively. Empirical coverage of the nominal 90% credible interval is 0.883 for both methods in Model 1, and is 0.887 for ILSA and 0.910 for NLSA in Model 2. The posterior-SD error is similar for the two methods in Model 1 and is 20.9% smaller for NLSA in Model 2. Figure 4 in the appendix gives the parameter-specific posterior results, and Table 5 summarizes estimation of posterior uncertainty. Thus the improvement in score estimation is also observed after posterior inference, without reducing interval coverage relative to ILSA.

Table 3: Posterior accuracy. MSE compares the approximate and exact posterior means; W_1 compares the full posteriors. Entries are mean $\pm$ sample SD across five training repetitions; lower is better. Bold and underline mark the best and second-best methods within each model and metric.

Method	Model 1		Model 2	
	Posterior-mean MSE ($\times 10^{-4}$)	W_1	Posterior-mean MSE ($\times 10^{-4}$)	W_1
Pilot-only NPE	7.900 $\pm$ 0.319	0.02251 $\pm$ 0.00061	47.061 $\pm$ 0.936	0.05274 $\pm$ 0.00072
Raw-data NPE	11.639 $\pm$ 2.760	0.02834 $\pm$ 0.00300	101.910 $\pm$ 8.305	0.08725 $\pm$ 0.00492
Raw-data FSM	7.200 $\pm$ 0.173	0.02142 $\pm$ 0.00048	46.709 $\pm$ 1.004	0.05270 $\pm$ 0.00055
ILSA	1.512 $\pm$ 0.222	0.01077 $\pm$ 0.00093	4.728 $\pm$ 0.225	0.01779 $\pm$ 0.00065
NLSA	0.804 $\pm$ 0.154	0.00811 $\pm$ 0.00088	3.078 $\pm$ 0.305	0.01438 $\pm$ 0.00113

4.3 Smith max-stable model

The Smith model represents the dependence between spatial extremes as a function of distance and direction. Let $h \in \mathbb{R}^2$ denote a spatial displacement and let $\delta_\Sigma(h)$ be the extremal coefficient, which is determined by the 2×2 storm-shape covariance matrix Σ . Each dataset contains 47 simulated fields of annual maxima at 79 Swiss stations, and the parameters of interest are the three distinct entries $(\Sigma_{11}, \Sigma_{12}, \Sigma_{22})$ of Σ . We consider three generating covariances: Low (234.1, -10.1 , 130.1), Center (332.2, 70.4, 184.6), and High (471.3, 199.1, 262.0), where the entries are given in the order $(\Sigma_{11}, \Sigma_{12}, \Sigma_{22})$. Center is the rainfall-reference covariance, while Low and High are prespecified departures from it. Computation uses the bounded positive-definite parameterization described in Appendix J, while all reported errors are evaluated on the covariance scale.

Although the full 79-site likelihood is unavailable, the exact bivariate likelihood is tractable for every pair of sites. Each field therefore gives 3,081 bivariate score contributions. We assign these contributions to 20 fixed groups defined by direction and distance and pool them before applying a common readout. ILSA pools the original scores, whereas NLSA transforms each score before the same pooling operation. This example thus compares the two methods when only local likelihood information is available. The covariance map, bivariate likelihood, extremal coefficient, grouping, and pilot are described in Appendix J.

Evaluation. Neither the exact full score nor the exact posterior is available for the Smith model. We therefore compare estimated quantities with their known values under the simulator. The main measure is the normalized integrated squared error (ISE) of the extremal coefficient over a common grid of spatial displacements. We also report separate MSEs for the three covariance coordinates by comparing the posterior-mean covariance from each neural method, or the MPLE, with the generating covariance. Smaller values are preferable for all these measures. Formal definitions and details of aggregation are given in Appendix L.

Extremal-dependence recovery. Table 4 shows that, compared with ILSA, NLSA reduces the normalized extremal-coefficient ISE by 16.2% at Low, 14.0% at Center, and 18.2% at High. Because the extremal coefficient depends jointly on the three covariance coordinates, this measure assesses recovery of the spatial dependence pattern rather than of the individual coordinates alone.

Figure 2 shows the distribution of this error over the two-dimensional lag plane. We

report all three parameter values because they represent different regimes of spatial dependence, rather than merely different levels of estimation difficulty.

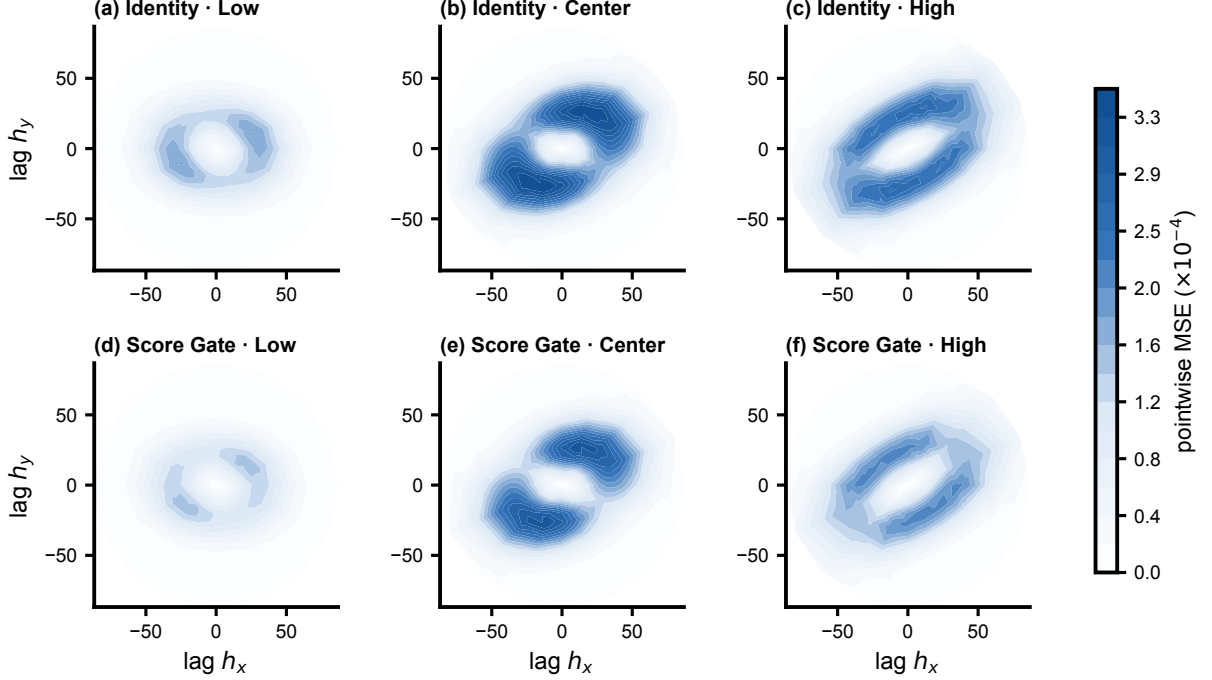


Figure 2: Pointwise MSE of the estimated Smith extremal coefficient over the lag plane, computed from 100 paired datasets at each parameter value. The top row shows ILSA and the bottom row shows NLSA. All panels use the same colour scale, with lighter blue indicating smaller error.

Covariance-parameter recovery. The coordinate-specific covariance errors give further information about the improvement in the estimated dependence surface. At Low, the largest reductions from ILSA to NLSA are for Σ_{11} (35.7%) and Σ_{22} (33.9%), whereas Σ_{12} improves by 4.5%. At Center, the reductions are 10.1%, 9.5%, and 9.9% for Σ_{11} , Σ_{12} , and Σ_{22} , respectively. At High, the largest reductions shift to Σ_{12} (36.3%) and Σ_{22} (34.1%), while Σ_{11} improves by 3.8%. A joint covariance measure would conceal these differences. Because the three coordinates have different numerical scales, we make method comparisons only within the same coordinate.

Table 4: Smith recovery across three fixed-truth regimes for all six methods. Each value uses 100 paired evaluation datasets, and smaller values are preferable. Bold and underlined entries are the best and second-best values within each parameter value and metric. The shaded [†] row marks the centre-specific advantage of Raw-data NPE caused by regression towards the centre of the prior.

Generating value	Method	Normalized extremal ISE ($\times 10^{-5}$)	Σ_{11} MSE	Σ_{12} MSE	Σ_{22} MSE
Low	All-pair MPLE	36.66	776.3	206.0	287.0
	Pilot-only NPE	38.77	1027.1	136.0	307.8
	Raw-data NPE	359.31	11669.0	6436.4	3738.7
	Raw-data FSM	23.15	547.5	72.3	222.6
	ILSA	<u>6.97</u>	<u>214.8</u>	<u>41.9</u>	<u>55.7</u>
	NLSA	5.84	138.2	40.0	36.8
Center	All-pair MPLE	58.67	1861.0	474.6	814.9
	Raw-data NPE [†]	6.63	139.9	22.5	53.2
	Pilot-only NPE	61.35	2030.6	563.5	796.9
	Raw-data FSM	47.12	1519.0	272.3	615.2
	ILSA	14.12	518.3	137.5	193.8
	NLSA	<u>12.14</u>	<u>466.0</u>	<u>124.4</u>	<u>174.6</u>
High	All-pair MPLE	48.98	2649.0	1037.9	917.6
	Pilot-only NPE	38.46	2123.9	1191.4	690.6
	Raw-data NPE	286.58	16812.5	16514.8	5022.2
	Raw-data FSM	21.45	1370.3	519.2	361.6
	ILSA	<u>12.35</u>	<u>577.3</u>	<u>259.8</u>	<u>255.1</u>
	NLSA	10.10	555.1	165.6	168.1

Other comparisons. The all-pair MPLE is a classical point-estimation benchmark; its extremal ISE is computed by substituting the estimate into the extremal-dependence surface. The raw-data methods provide more general comparisons with methods that do not use the local-score representation, but they do not isolate the effect of NLSA. In this experiment, the posterior means from Raw-data NPE remain close to the centre of the prior at all three parameter values. It consequently overestimates at Low and underestimates at High. Its small error at Center should not be viewed separately from these results, because Center coincides with the point towards which its estimates regress. Figure 5 in the appendix shows the complete distributions of posterior-mean errors. Further details of the baselines and MPLE are given in Appendices G and J.

Pilot-only NPE is less accurate than ILSA and NLSA on every reported Smith metric under all three regimes. This comparison shows that the learned score field contributes substantial information beyond the pilot, although it changes the Stage-2 summary and is therefore not a matched test of the nonlinear pre-pooling map.

Table 6 in the appendix shows that the empirical coverage of 90% credible intervals is similar for NLSA and ILSA across the three parameter values and covariance coordinates.

Summary. In summary, Models 1 and 2 show that transforming the local scores before pooling improves both estimation of the exact score and approximation of the exact posterior. In the Smith model, where neither exact reference is available, it also improves estimation of the extremal-dependence surface and of the covariance parameters that determine it. The comparison between ILSA and NLSA supports attribution of these improvements to the local transformation; the remaining baselines provide comparisons with more general alternatives.

5 Discussion

The controlled examples compare ILSA and NLSA within an otherwise fixed architecture. They show that the reduction in score-estimation error is accompanied by a reduction in posterior-approximation error. The Smith example gives a similar comparison for a spatial extremes model whose full likelihood and exact posterior cannot be evaluated.

There are several limitations to the current numerical evidence. First, the controlled examples use particular block structures, while the raw-data baseline uses a generic architecture. Second, Stage 1 for the Smith model is trained only once. Third, we do not investigate sensitivity to the tube width. Finally, we do not compare conditioning on the pilot-score pair with conditioning on the one-step estimator or on a score evaluated at the true parameter. The present empirical conclusions should therefore be restricted to the matched comparison between ILSA and NLSA in the settings considered.

6 Conclusion

This work proposes a method for using tractable local likelihood scores in SBI when the joint likelihood is intractable. The local scores are transformed before they are pooled within homogeneous groups, and tube Fisher-score matching is used to estimate the resulting score function without evaluating the joint likelihood. Posterior inference then conditions jointly on a pilot estimate and the estimated score at the pilot. In two controlled models, NLSA improves estimation of the exact score and approximation of the exact posterior relative to pooling the original scores. It also improves estimation of extremal dependence in the Smith max-stable example.

Appendix

A Motivating examples

The first example shows that nonlinear aggregation can improve on the optimal linear composite score even when the required information is contained in marginal and pairwise likelihoods. The second example shows that the nonlinear transformation may need to be applied before pooling. We present only the calculations required to establish these points.

Elliptical field with banded precision. Let $X = (X_1, \dots, X_n)^\top$ have a multivariate Student- t density with degrees of freedom $\nu > 0$ and scatter matrix $\Sigma(\rho, \sigma^2)$ whose inverse is tridiagonal, where σ^2 is the innovation-scale parameter in this precision parametrization. Define

$$S_0 = \sum_{j=1}^n X_j^2, \quad S_1 = \sum_{j=1}^{n-1} X_j X_{j+1}, \quad E = X_1^2 + X_n^2,$$

where E collects the boundary terms. Then

$$Q(X) = X^\top \Sigma^{-1} X = \sigma^{-2} \{(1 + \rho^2)S_0 - \rho^2 E - 2\rho S_1\},$$

and the full score is a nonlinear function

$$\nabla_{(\rho, \sigma^2, \nu)} \log p(X \mid \rho, \sigma^2, \nu) = \kappa_{\rho, \sigma^2, \nu}(S_0, S_1, E),$$

because the Student- t log likelihood contains $(\nu + Q)^{-1}$ and $\log(1 + Q/\nu)$. In the Gaussian model with the same precision structure, κ is affine and linear composite aggregation is exact.

For a d -dimensional Student- t margin with density f_d and Mahalanobis statistic Q_d , define its log-scale score by

$$s_d := \partial_{\log \sigma} \log f_d = -d + (\nu + d) \frac{Q_d}{\nu + Q_d} \in (-d, \nu).$$

The inverse link is

$$\phi_d(s) = \nu \frac{s + d}{\nu - s}, \quad \text{so that} \quad \phi_d(s_d) = Q_d.$$

For lag-one pair j , set $P_j = X_j^2 + X_{j+1}^2$ and $C_j = X_j X_{j+1}$. After recovering Q_{2j} from the pair log-scale score, an affine rescaling $B_{\rho, j}$ of the pair dependence-score coordinate gives

$$\sigma^2(1 - \rho^2)Q_{2j} = P_j - 2\rho C_j, \quad B_{\rho, j} = \rho P_j - (1 + \rho^2)C_j.$$

The coefficient matrix for (P_j, C_j) therefore has determinant $-(1 - \rho^2)$ and remains invertible for interior $|\rho| < 1$; Appendix E.1 gives the rescaling $B_{\rho, j}$. Summing the recovered C_j gives S_1 , while the P_j and marginal terms give S_0 . Exact recovery also requires the pooling layout to expose the two boundary coordinates entering E . Thus the grouping and positional classes remain important even when each local score has an inverse transformation. The inverse $\phi_d(s)$ diverges as $s \uparrow \nu$, which suggests that the learned transformation should not be bounded. This example demonstrates that

nonlinear aggregation can recover Fisher-score information outside the optimal linear span of the composite scores.

Independent latent blocks. Now consider K independent blocks, each containing m observations,

$$\zeta_b \sim \text{Bernoulli}(\pi), \quad X_{bj} = \varepsilon_{bj} + \zeta_b \tau, \quad \varepsilon_{bj} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

where $b = 1, \dots, K$, $j = 1, \dots, m$, $\tau > 0$ is fixed, and $\pi \in (0, 1)$ is the parameter of interest. Let $\text{expit}(a) = (1 + e^{-a})^{-1}$, $\beta = \text{logit}(\pi)$, $\ell(x) = \tau x - \tau^2/2$, and

$$h_\beta(a) = \text{expit}(\beta + a) - \text{expit}(\beta).$$

The one-observation score $s_{bj} = h_\beta\{\ell(X_{bj})\}$ lies in $(-\pi, 1 - \pi)$ and has inverse

$$\phi_\beta(s) = \text{logit}(\pi + s) - \beta, \quad \text{so that} \quad \phi_\beta(s_{bj}) = \ell(X_{bj}).$$

Thus the transformed within-block sum recovers

$$L_b = \sum_{j=1}^m \ell(X_{bj}), \quad S_\beta(X) = \sum_{b=1}^K h_\beta(L_b).$$

This failure of representation is exact and does not require a central limit theorem. Because h_β is strictly non-affine, for every $m \geq 2$ there are two blocks with the same pooled mean $m^{-1} \sum_j h_\beta(\ell_j)$ but different values of $\sum_j \ell_j$. Otherwise, $\sum_j \ell_j$ would be a function of $\sum_j h_\beta(\ell_j)$ for all inputs, and differentiation with respect to each ℓ_i would imply that h'_β is constant. It follows that a block readout applied after linear pooling cannot distinguish all such blocks. Exact recovery instead requires the order

$$\begin{aligned} \text{local transform} &\rightarrow \text{within-block pool} \rightarrow \text{block nonlinearity} \\ &\rightarrow \text{block sum.} \end{aligned}$$

Thus, within a latent block, the nonlinear transformation must precede a pooling operation that would otherwise remove the information required to construct the block score.

B Algorithm

Algorithm 1 NLSA Pilot-Score NPE

Require: Simulator $p(\cdot \mid \theta)$; prior Π_0 ; local densities $\{f_j(\cdot \mid \beta) : j \in \mathcal{J}\}$ with groups $\mathcal{U}_1, \dots, \mathcal{U}_{G_{\text{grp}}}$; anchor distribution λ_{anc} ; tube kernel $q_{\text{tube}}(\theta \mid \beta) = \mathcal{N}_p(\theta; \beta, \Sigma_q)$; pilot estimator $\hat{\beta}(\cdot)$; NLSA surrogate score S_ω of Eq. (2) with the map in Eq. (??); NPE family $q_\lambda(\theta \mid c)$.

Stage 1: tube FSM for the surrogate score

- 1: Initialize $\omega = (\psi, \eta)$ at ILSA, i.e. $m_{\eta, g} \equiv 0$ for all g
- 2: **repeat**
- 3: Draw a minibatch of $\beta \sim \lambda_{\text{anc}}, \theta \sim q_{\text{tube}}(\cdot \mid \beta), X \sim p(\cdot \mid \theta)$
- 4: Step on ω to reduce $\|S_\omega(X, \beta) - \Sigma_q^{-1}(\theta - \beta)\|_2^2$
- 5: **until** converged
- 6: Freeze $\hat{\omega} \leftarrow \omega$

Stage 2: NPE on the pilot-score summary

- 7: **repeat**
- 8: Draw a minibatch of $\theta \sim \Pi_0, X \sim p(\cdot \mid \theta)$
- 9: Form $c(X) = \{\hat{\beta}(X), S_{\hat{\omega}}(X, \hat{\beta}(X))\}$ as in Eq. (7)
- 10: Step on λ to reduce $-\log q_\lambda\{\theta \mid c(X)\}$
- 11: **until** converged
- 12: Set $\hat{\lambda} \leftarrow \lambda$

Inference

- 13: **return** $q_{\hat{\lambda}}\{\theta \mid c(X_{\text{obs}})\}$, or samples from it, with $c(X_{\text{obs}})$ formed as above
-

C Information retained by a pooled representation.

Let $C_n \in \mathbb{R}^{d_C}$ denote a centered, $n^{1/2}$ -scaled linearly pooled vector of fixed dimension. For a local displacement $\alpha \in \mathbb{R}^p$, let

$$\theta_n(\alpha) = \theta_0 + \alpha/\sqrt{n}.$$

Corollary 5 (Gaussian pooled representations). Assume that the local experiments generated by C_n under $\theta_n(\alpha)$ converge to

$$C_\alpha \sim \mathcal{N}_{d_C}(\Lambda\alpha, \Sigma_C),$$

where $\Lambda \in \mathbb{R}^{d_C \times p}$ is the local mean-sensitivity matrix and $\Sigma_C \in \mathbb{R}^{d_C \times d_C}$ is positive definite. Then the limiting experiment has Fisher information

$$G_C = \Lambda^\top \Sigma_C^{-1} \Lambda,$$

which is attained by the linear statistic

$$Z_C = \Lambda^\top \Sigma_C^{-1} C_\alpha.$$

Consequently, every measurable outer readout of C_α has Fisher information no greater than G_C .

When C_n is formed from normalized sums of local scores, Z_C corresponds to the Godambe-optimal linear weighting of those pooled scores [Lindsay, 1988, Lindsay et al.,

2011, Varin et al., 2011]. Thus Corollary 5 formalizes the post-pooling information ceiling described above.

Proof. Apply the information-bound argument of Proposition 1 to the limiting Gaussian experiment. Let $C \sim \mathcal{N}_{d_C}(\Lambda\alpha, \Sigma_C)$. Relative to $\alpha = 0$, its log-likelihood ratio is

$$\log \frac{dQ_\alpha}{dQ_0}(C) = \alpha^\top \Lambda^\top \Sigma_C^{-1} C - \frac{1}{2} \alpha^\top \Lambda^\top \Sigma_C^{-1} \Lambda \alpha,$$

where Q_α denotes the law of C under local displacement α . Hence

$$Z_C = \Lambda^\top \Sigma_C^{-1} C$$

is sufficient for α in the Gaussian limit, has covariance $G_C = \Lambda^\top \Sigma_C^{-1} \Lambda$ at $\alpha = 0$, and is a linear function of C . If $R = r(C)$ is any measurable outer readout, the score of the induced experiment based on R is $\mathbb{E}_0(Z_C \mid R)$, so its information is

$$\text{Var}_0\{\mathbb{E}_0(Z_C \mid R)\} \preceq \text{Var}_0(Z_C) = G_C$$

by the conditional-variance decomposition.

The remaining statement transfers this limit-experiment result to sequences of outer readouts. The assumed weak convergence and the asymptotic representation theorem place every tight weak subsequential limit inside the Gaussian shift experiment. This conclusion requires the standard regularity conditions on the decision problem. Since every sample-size-dependent decision rule δ_n based on an outer readout of C_n is measurable with respect to C_n , the same representation applies to such readouts. The local asymptotic minimax lower bound therefore equals the Gaussian-shift benchmark. When G_C is positive definite, the convolution theorem gives the regular-estimator specialization. Any regular limit law has an efficient Gaussian component with covariance G_C^{-1} plus an independent residual component. Its covariance is therefore at least G_C^{-1} whenever finite second moments exist. Finally,

$$G_C^{-1} Z_C \sim \mathcal{N}_p(\alpha, G_C^{-1})$$

in the limiting experiment, proving attainability of this covariance bound. □

D Population target of tube FSM

Let λ_β be a probability distribution over the evaluation points $\beta \in \Theta$, and let $q_{\text{tube}}(\theta \mid \beta) = N_p(\theta; \beta, \Sigma_q)$, where $\Sigma_q \in \mathbb{R}^{p \times p}$ is positive definite and controls the tube width. We draw $\beta \sim \lambda_\beta$, $\theta \sim q_{\text{tube}}(\cdot \mid \beta)$, and $X \sim p(\cdot \mid \theta)$, and minimize

$$\mathcal{L}_{\text{FSM}}(\omega) = \mathbb{E}[\|S_\omega(X, \beta) - \nabla_\beta \log q_{\text{tube}}(\theta \mid \beta)\|_2^2]. \quad (8)$$

The regression response is evaluable because $\nabla_\beta \log q_{\text{tube}}(\theta \mid \beta) = \Sigma_q^{-1}(\theta - \beta)$. Hence (8) requires simulation on the ‘tube’ of (θ, β) but not evaluation of the joint likelihood, the width along the diagonal $\theta = \beta$ is determined by the covariance matrix Σ_q .

Define the corresponding smoothed likelihood and local posterior density by

$$p_{\text{tube}}(x \mid \beta) = \int p(x \mid \theta) q_{\text{tube}}(\theta \mid \beta) d\theta,$$

$$P_{\text{tube}}(\theta \mid x, \beta) = \frac{p(x \mid \theta) q_{\text{tube}}(\theta \mid \beta)}{p_{\text{tube}}(x \mid \beta)}.$$

The tube construction makes the joint distribution of (X, β) during training closer to that encountered when the method is applied, for which the input is

$$(X_{\text{obs}}, \hat{\beta}(X_{\text{obs}})).$$

If training were restricted to $\beta = \theta$, the method would observe only local scores evaluated at the generating parameter. In contrast, evaluation at a data-dependent pilot usually produces shifted local scores. Sampling θ in a neighbourhood of β exposes the estimated score function to such off-diagonal pairs during training.

Proposition 6. Assume that differentiation under the integral sign is valid, that integration by parts in θ has vanishing boundary terms, and that S_ω , $\nabla_\beta \log q_{\text{tube}}$, and g_0 are square-integrable. Then the following four criteria differ by additive constants that do not depend on ω , and therefore have the same minimizers:

$$\begin{aligned} & \arg \min_{\omega} \mathbb{E}_{x \sim p(\cdot \mid \theta), \theta \sim q_{\text{tube}}(\cdot \mid \beta), \beta \sim \lambda_{\text{anc}}(\cdot)} \left[\|S_\omega(x, \beta) - \nabla_\beta \log q_{\text{tube}}(\theta \mid \beta)\|^2 \right] \\ &= \arg \min_{\omega} \mathbb{E}_{x \sim p_{\text{tube}}(\cdot \mid \beta), \beta \sim \lambda_{\text{anc}}(\cdot)} \left[\|S_\omega(x, \beta) - \nabla_\beta \log p_{\text{tube}}(x \mid \beta)\|^2 \right] \\ &= \arg \min_{\omega} \mathbb{E}_{x \sim p(\cdot \mid \theta), \theta \sim q_{\text{tube}}(\cdot \mid \beta), \beta \sim \lambda_{\text{anc}}(\cdot)} \left[\|S_\omega(x, \beta) - g_0(x, \theta)\|^2 \right] \\ &= \arg \min_{\omega} \mathbb{E}_{x \sim p_{\text{tube}}(\cdot \mid \beta), \beta \sim \lambda_{\text{anc}}(\cdot)} \left[\|S_\omega(x, \beta) - \mathbb{E}_{\theta \sim P_{\text{tube}}(\cdot \mid x, \beta)} \{g_0(x, \theta) \mid x, \beta\}\|^2 \right]. \end{aligned}$$

Proposition 6 shows that the population regression target is the Fisher score of the smoothed likelihood. Under the stated regularity conditions, this target approaches the desired score function as $\Sigma_q \rightarrow 0$. Equivalently, it is the conditional expectation of the exact Fisher score under the local posterior $P_{\text{tube}}(\theta \mid x, \beta)$.

Proof of Proposition 6. All four criteria have the same $\mathbb{E}\|S_\omega\|^2$ term, so it is sufficient to compare their cross terms.

First equality. The cross term of the smoothed-score criterion is

$$\begin{aligned} & \mathbb{E}_{x \sim p(\cdot \mid \theta), \theta \sim q_{\text{tube}}(\cdot \mid \beta), \beta \sim \lambda_{\text{anc}}(\cdot)} \{S_\omega(x, \beta)^\top \nabla_\beta \log p_{\text{tube}}(x \mid \beta)\} \\ &= \int S_\omega(x, \beta)^\top \frac{\int p(x \mid \theta) \nabla_\beta q_{\text{tube}}(\theta \mid \beta) d\theta}{\int p(x \mid \theta) q_{\text{tube}}(\theta \mid \beta) d\theta} \left[\int p(x \mid \theta) q_{\text{tube}}(\theta \mid \beta) d\theta \right] \lambda_{\text{anc}}(\beta) dx d\beta \\ &= \int S_\omega(x, \beta)^\top \nabla_\beta \log q_{\text{tube}}(\theta \mid \beta) q_{\text{tube}}(\theta \mid \beta) p(x \mid \theta) \lambda_{\text{anc}}(\beta) dx d\theta d\beta, \end{aligned}$$

which is the cross term of the second criterion.

Second equality. The cross term with the exact score at the generating parameter is

$$\begin{aligned}
& \mathbb{E}_{(x,\theta,\beta)} \{S_\omega(x, \beta)^\top g_0(x, \theta)\} \\
&= \int S_\omega(x, \beta)^\top \nabla_\theta \log p(x | \theta) p(x | \theta) q_{\text{tube}}(\theta | \beta) \lambda_{\text{anc}}(\beta) dx d\theta d\beta \\
&= \int S_\omega(x, \beta)^\top \int \nabla_\theta \log p(x | \theta) \frac{p(x | \theta) q_{\text{tube}}(\theta | \beta)}{p_{\text{tube}}(x | \beta)} d\theta p_{\text{tube}}(x | \beta) \lambda_{\text{anc}}(\beta) dx d\beta \\
&= \int S_\omega(x, \beta)^\top \int \nabla_\theta p(x | \theta) q_{\text{tube}}(\theta | \beta) d\theta \lambda_{\text{anc}}(\beta) dx d\beta \\
&= - \int S_\omega(x, \beta)^\top \int p(x | \theta) \nabla_\theta q_{\text{tube}}(\theta | \beta) d\theta \lambda_{\text{anc}}(\beta) dx d\beta \\
&= - \int S_\omega(x, \beta)^\top \nabla_\theta \log q_{\text{tube}}(\theta | \beta) q_{\text{tube}}(\theta | \beta) p(x | \theta) \lambda_{\text{anc}}(\beta) dx d\theta d\beta \\
&= - \mathbb{E}_{(x,\theta,\beta)} \{S_\omega(x, \beta)^\top \nabla_\theta \log q_{\text{tube}}(\theta | \beta)\} \\
&= \mathbb{E}_{(x,\theta,\beta)} \{S_\omega(x, \beta)^\top \nabla_\beta \log q_{\text{tube}}(\theta | \beta)\},
\end{aligned}$$

where the last equality uses $\nabla_\theta \log q_{\text{tube}}(\theta | \beta) = -\nabla_\beta \log q_{\text{tube}}(\theta | \beta)$. Thus this criterion has the same cross term as the first two criteria.

Third equality. Since $S_\omega(x, \beta)$ is measurable with respect to (x, β) , the tower property gives

$$\mathbb{E}\{S_\omega(x, \beta)^\top g_0(x, \theta)\} = \mathbb{E}\left[S_\omega(x, \beta)^\top \mathbb{E}_{\theta \sim P_{\text{tube}}(\cdot | x, \beta)}\{g_0(x, \theta) | x, \beta\}\right],$$

which is the cross term of the last criterion. $\square$

E Technical proofs

E.1 Elliptical pair-map calculation

For a standardized lag-one Student- t pair with scatter correlation ρ ,

$$Q_{2j} = \frac{P_j - 2\rho C_j}{\sigma^2(1 - \rho^2)}, \quad P_j = X_j^2 + X_{j+1}^2, \quad C_j = X_j X_{j+1}.$$

Let $s_{\rho,j} = \partial_\rho \log f_2(X_j, X_{j+1})$ be the pair dependence score. Direct differentiation gives

$$s_{\rho,j} = \frac{\rho}{1 - \rho^2} - \frac{\nu + 2}{\sigma^2(1 - \rho^2)^2(\nu + Q_{2j})} \{\rho P_j - (1 + \rho^2)C_j\}.$$

Hence, after recovering Q_{2j} from the log-scale score, define the observable rescaling

$$B_{\rho,j} := \frac{\sigma^2(1 - \rho^2)^2(\nu + Q_{2j})}{\nu + 2} \left\{ \frac{\rho}{1 - \rho^2} - s_{\rho,j} \right\}.$$

Then $B_{\rho,j} = \rho P_j - (1 + \rho^2)C_j$. Together with $\sigma^2(1 - \rho^2)Q_{2j} = P_j - 2\rho C_j$, the coefficient matrix is

$$\begin{pmatrix} 1 & -2\rho \\ \rho & -(1 + \rho^2) \end{pmatrix},$$

whose determinant is $-(1 - \rho^2)$.

E.2 Proof of Proposition 1

Fix a measurable representation $T = T(U)$ and a centered estimating function $f = f(T)$. Because f is $\sigma(T)$ -measurable,

$$H_f = \mathbb{E}_{\theta_0}\{f g_0^\top\} = \mathbb{E}_{\theta_0}\{f m_T(T)^\top\}.$$

The best linear predictor of $m_T(T)$ from f has coefficient matrix $H_f^\top V_f^{-1}$. Its residual covariance is therefore

$$I_T - H_f^\top V_f^{-1} H_f = I_T - G(f) \succeq 0,$$

which proves $G(f) \preceq I_T$ and gives equality at $f = m_T$ when I_T is nonsingular.

Since T is a measurable function of U , the tower property gives

$$m_T(T) = \mathbb{E}_{\theta_0}\{m_0(U) \mid T\}.$$

Hence

$$m_0(U) = m_T(T) + \{m_0(U) - m_T(T)\},$$

where the second term has conditional mean zero given T and is therefore orthogonal to $m_T(T)$. Thus

$$I_U = I_T + \text{Var}_{\theta_0}\{m_0(U) - m_T(T)\},$$

which proves $I_T \preceq I_U$ and the stated matrix-valued coarsening identity. The covariance term vanishes exactly when $m_0(U) = m_T(T)$ almost surely, equivalently when $m_0(U)$ is $\sigma(T)$ -measurable. Applying the same conditional-variance decomposition to the full Fisher score gives

$$I(\theta_0) = I_U + \mathbb{E}_{\theta_0}\{\text{Var}(g_0 \mid U)\},$$

so $I_U \preceq I(\theta_0)$.

For every scalar $\ell \in \mathbb{L}$, ℓ is $\sigma(U)$ -measurable, hence

$$\mathbb{E}_{\theta_0}\{(g_0 - m_0)\ell\} = 0.$$

Thus the coordinatewise projections of g_0 and m_0 onto $\mathbb{L}$ agree. Write $g_L = \Pi_{\mathbb{L}} g_0 = \Pi_{\mathbb{L}} m_0$. Orthogonality gives

$$\mathbb{E}(g_L g_0^\top) = \mathbb{E}(g_L g_L^\top) = \text{Var}(g_L),$$

so, when this covariance is nonsingular, $G(g_L) = \text{Var}(g_L)$. The same linear-prediction argument applied within the closed linear span shows that no linear composite score has larger Godambe information. Finally,

$$I_U = \text{Var}(g_L) + \text{Var}(m_0 - g_L),$$

which proves the displayed nonlinear-versus-linear gap identity and the stated strictness criterion.

E.3 Proof of Proposition 3

Under the local translation model, an evaluation-point perturbation gives $d\Psi = -H_0 d\beta$. By the chain rule,

$$dF = -F_\Psi H_0 d\beta.$$

First-order pilot invariance for every $d\beta$ therefore implies $F_\Psi H_0 = 0$. Since H_0 is non-singular, $F_\Psi = 0$. A generating-parameter perturbation gives $d\Psi = H_0 d\theta$, so $dF = F_\Psi H_0 d\theta = 0$. Thus first-order pilot invariance forces loss of first-order information about θ .

E.4 Proof of Proposition 4

For $F(\beta, \Psi)$, an evaluation-point perturbation yields

$$dF = (F_\beta - F_\Psi H_0) d\beta.$$

Hence invariance for every $d\beta$ is equivalent to $F_\beta = F_\Psi H_0$. Define the characteristic coordinate $z = \beta + H_0^{-1} \Psi$. Along a curve satisfying $dz = 0$, we have $d\Psi = -H_0 d\beta$, and the transport equation gives $dF = 0$. Thus F remains locally constant on level sets of z and takes the form $F(\beta, \Psi) = \Xi(z)$ for a differentiable Ξ . Conversely, direct differentiation shows that every such composition satisfies the transport equation.

F Notation table

Symbol	Meaning
θ	generating parameter for a simulated or hypothetical dataset
β	score-evaluation point
θ_0	true parameter in local asymptotics
p	parameter dimension
n	effective sample size in local asymptotics
$s_j(X; \beta)$	tractable local likelihood score, a vector in $\mathbb{R}^p$
U	complete whole-sample collection of local scores at the comparison point
$m_0(U)$	conditional Fisher score $\mathbb{E}(g_0 U)$ and local score-information-optimal target
$T = T(U)$	generic measurable representation retained after a possible coarsening of U
$m_T(T)$	conditional Fisher score $\mathbb{E}(g_0 T) = \mathbb{E}\{m_0(U) T\}$
I_T	information $\text{Var}\{m_T(T)\}$ retained by representation T
$\mathbb{L}$	closed scalar linear span of all local-score coordinates
I_U	information $\text{Var}\{m_0(U)\}$ retained by the complete local score sigma-field
C_n	centered, $n^{1/2}$ -scaled linearly pooled vector in the local experiment
G_C	information $\Lambda^\top \Sigma_C^{-1} \Lambda$ of the Gaussian limit of C_n
U_b	full collection of local scores retained before pooling in readout block b
g_b	exact Fisher-score contribution associated with readout block b ; for $K = 1$, $g_1 = g_0$
$m_b(U_b; \beta)$	conditional block-score target $\mathbb{E}_\beta\{g_b(X, \beta) U_b\}$
$\mathcal{U}_g$	group of local components with shared structural role
$b = 1, \dots, K$	outer-block index and number of outer blocks
R_{gb}	pooled transformed feature for group g in block b
$\phi_{\eta, g}$	learned local transformation for group g
$\mathcal{R}_\psi$	shared nonlinear block readout
$S_{\psi, \eta}$	learned structured score surrogate
$\omega = (\psi, \eta)$	combined parameter vector for the structured score surrogate
λ_{anc}	probability distribution of the tube anchor β
Σ_q	positive-definite covariance matrix controlling the tube width
$q_{\text{tube}}(\theta \beta)$	Gaussian tube density with mean β and covariance Σ_q
$p_{\text{tube}}(x \beta)$	likelihood obtained by averaging $p(x \theta)$ under $q_{\text{tube}}(\theta \beta)$
$P_{\text{tube}}(\theta x, \beta)$	local posterior induced by $p(x \theta) q_{\text{tube}}(\theta \beta)$
Ψ	centered estimating function derived from $S_{\psi, \eta}$
$H(\beta)$	sensitivity matrix of Ψ at evaluation point β
$V(\beta)$	variability matrix of Ψ at evaluation point β
$G(\beta)$	Godambe information $H^\top V^{-1} H$
Z_{pair}	pilot-score summary used for posterior inference

G Additional baselines

The additional baselines assess the contributions of different parts of the two-stage method. They change the Stage-2 summary, the input representation, or the Stage-1

architecture. They therefore provide general comparisons, whereas ILSA and NLSA give the direct comparison of interest.

Pilot-only NPE. For all three models, Pilot-only NPE assesses whether the estimated score contains information beyond the data-derived pilot. It omits the fixed Stage-1 score and conditions only on

$$c_{\text{pilot}}(Y) = \hat{u}(Y). \quad (9)$$

The corresponding posterior estimator is

$$q_{\lambda}^{\text{pilot}}(\theta \mid \hat{u}(Y)). \quad (10)$$

This baseline uses the same pilot, Stage-2 simulation budget, neural posterior estimation (NPE) family, and training procedure. Its comparison with the score-based methods measures the additional posterior information provided by the estimated score.

Raw-data NPE. Raw-data NPE omits both the pilot and Stage 1 and performs posterior inference directly from the observations. Writing

$$x_{\text{raw}}(Y) = \text{vec}(Y), \quad (11)$$

the posterior estimator is

$$q_{\lambda}^{\text{raw}}(\theta \mid x_{\text{raw}}(Y)). \quad (12)$$

For Models 1 and 2, $x_{\text{raw}}(Y)$ contains respectively 400 and 1,600 flattened observation coordinates. For Smith, it contains the flattened 47×79 field. The score-based and raw-data approaches use the same mixture-density-network NPE (MDN-NPE) family and Stage-2 simulation budget, but their input dimensions differ. This baseline assesses whether the posterior can be estimated directly from the observations without using the pilot or local-score representation.

Raw-data FSM. Raw-data FSM retains the two-stage construction but replaces the tractable local-score representation with a score function estimated directly from raw observations. It uses the same anchored FSM target as in Section 3.3,

$$t = \Sigma_q^{-1}(\tilde{u} - u_0), \quad (13)$$

and predicts this target from Y while excluding the local score contributions $s_{kgj}(Y; u_0)$.

For Models 1 and 2, the Stage-1 field is an MLP acting on the flattened dataset and anchor,

$$\hat{S}_{\omega}^{\text{raw}}(Y, u_0) = g_{\omega}(\text{vec}(Y), u_0). \quad (14)$$

For Smith, writing $Y_t \in \mathbb{R}^{79}$ for the t th simulated annual-maxima field, we apply a shared MLP to each annual field and sum the resulting score contributions:

$$\hat{S}_{\omega}^{\text{raw}}(Y, u_0) = \sum_{t=1}^{47} g_{\omega}(Y_t, u_0). \quad (15)$$

After Stage 1, we fix the raw-data score function and use it in the same pilot-based

Stage-2 summary as the score-based methods:

$$c_{\text{rawFSM}}(Y) = \left(\hat{u}(Y), \hat{S}_{\text{frozen}}^{\text{raw}}(Y, \hat{u}(Y)) \right). \quad (16)$$

Thus Raw-data FSM preserves the two-stage FSM–NPE procedure but does not use either the tractable local-score representation or its transformation before pooling.

All-pair MPLE. For Smith, we additionally include the all-pair maximum pairwise-likelihood estimator (MPLE) as a classical point-estimation baseline. Let f_{ij} denote the exact Smith bivariate likelihood contribution for sites i and j . Using all station pairs and all 47 annual fields, the estimator is

$$\hat{u}_{\text{MPLE}}(Y) = \arg \max_u \sum_{t=1}^{47} \sum_{i < j} \log f_{ij}(Y_{ti}, Y_{tj}; r^{-1}(u)), \quad (17)$$

with covariance estimate

$$\hat{\Sigma}_{\text{MPLE}}(Y) = r^{-1}(\hat{u}_{\text{MPLE}}(Y)). \quad (18)$$

MPLE returns a point estimate rather than a posterior distribution. It provides a classical benchmark for estimation of the Smith covariance. The direct comparison of the two neural methods remains that between ILSA and NLSA. Appendix J gives the bivariate likelihood, covariance parameterization, and numerical optimization details.

Comparison scope. The baselines answer complementary questions. Pilot-only NPE measures the learned score’s contribution beyond the pilot. Raw-data NPE tests direct posterior inference from observations. Raw-data FSM retains both stages while replacing local scores with raw data. The Smith all-pair MPLE provides a classical pairwise-likelihood reference. Because these comparisons change the summary, representation, architecture, or inferential output, they do not isolate the NLSA transformation. Evidence about that transformation comes from the direct comparison between ILSA and NLSA in Section 4.1.

Tables 2 and 3 report the five-repetition controlled results for these methods. We report no additional single-run aggregate.

Additional Smith baselines. The Smith baselines use the same 100 paired datasets at each parameter value as Table 4. Raw-data NPE remains close to the centre of the prior at all three values. Its small error at Center therefore does not, by itself, demonstrate recovery from the data, because its estimates also shift towards Center at Low and High. Raw-data FSM responds to the data but has larger covariance MSE and normalized extremal-coefficient ISE than both local-score methods at every parameter value. These baselines change both the input representation and the architecture, so only the comparison between ILSA and NLSA isolates the transformation before pooling. Table 4 gives the aggregate results for all three parameter values, and Figure 5 gives the corresponding per-dataset error distributions.

H Detailed numerical results

This section provides detailed results complementing the main-text experiments. Models 1–2 use five Stage-1 training repetitions. We evaluate Stage-1 score accuracy on 5,000 held-out datasets at each generating value. We evaluate Stage-2 posterior accuracy on 10 paired held-out datasets per generating value and repetition.

H.1 Exact-score recovery in Models 1 and 2

Figure 3 disaggregates the main-text Stage-1 results by generating value. The paired repetition-level results show whether the aggregate score-recovery gain is consistent across the parameter range. For Model 1, the endpoint values $\pi = 0.07$ and 0.68 are additional held-out diagnostics from the same fixed checkpoints; the main-text aggregate remains the prespecified four-value comparison.

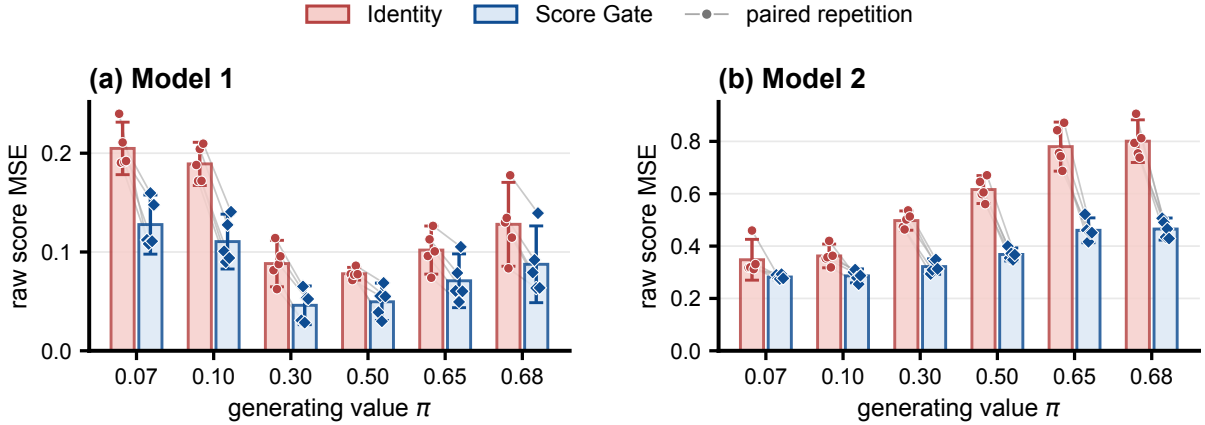


Figure 3: Per-parameter Stage-1 exact-score recovery for Models 1–2. Points denote the five Stage-1 training repetitions and summaries show the repetition mean with 95% t intervals. Both panels report raw score MSE. Compare methods at the same generating value; lower is better.

H.2 Posterior accuracy in Models 1 and 2

Figure 4 reports the two exact-reference posterior metrics used to assess approximation accuracy. Within each repetition and generating value, each method is first averaged over the same 10 paired held-out datasets.

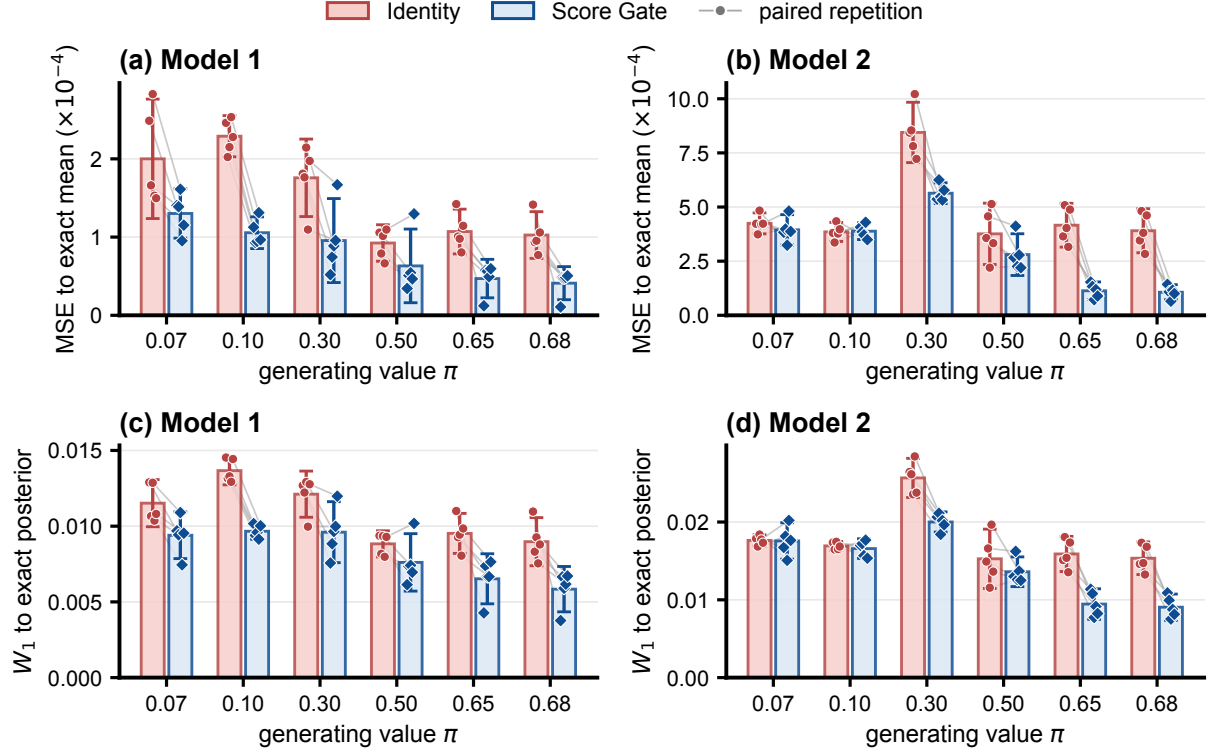


Figure 4: Posterior accuracy by parameter value for Models 1 and 2. The panels show ILSA and NLSA as absolute errors at each generating value; lower is better. Points denote the five Stage-1 training repetitions, while bars and error bars show the repetition mean and 95% t interval. The top row reports MSE to the exact posterior mean and the bottom row reports W_1 distance to the exact posterior. Gray segments join matched repetitions.

H.3 Posterior uncertainty in Models 1 and 2

Table 5: Aggregate posterior-uncertainty diagnostics for Models 1–2. SD error is the mean absolute difference from the exact posterior standard deviation; coverage refers to nominal 90% credible intervals.

Model	Exact mean SD	Mean posterior SD		SD error		Coverage 90%	
		ILSA	NLSA	ILSA	NLSA	ILSA	NLSA
Model 1	0.09695	0.09846	0.09776	0.004526	0.004564	0.883	0.883
Model 2	0.068088	0.072411	0.071373	0.006974	0.005518	0.887	0.910

H.4 Results for the Smith model

The main text reports aggregate errors for all three generating parameter values. Figure 5 shows the corresponding posterior-mean error distributions over the same 100 paired datasets per parameter value. Raw-data NPE posterior means remain concentrated near the centre of the bounded prior across Low, Center, and High. It therefore overestimates at Low and underestimates at High, but appears numerically strong at Center because this generating value coincides with the location towards which its estimates regress. Readers should therefore avoid interpreting its center ranking in isolation.

Table 6: Empirical coverage of central 90% Smith posterior intervals over the same 100 datasets at each generating parameter value. The nominal target is 0.90.

Generating value	Method	Σ_{11}	Σ_{12}	Σ_{22}
Low	ILSA	0.93	0.97	0.94
	NLSA	0.93	0.97	0.95
Center	ILSA	0.84	0.88	0.88
	NLSA	0.90	0.90	0.87
High	ILSA	0.95	0.94	0.94
	NLSA	0.94	0.96	0.96

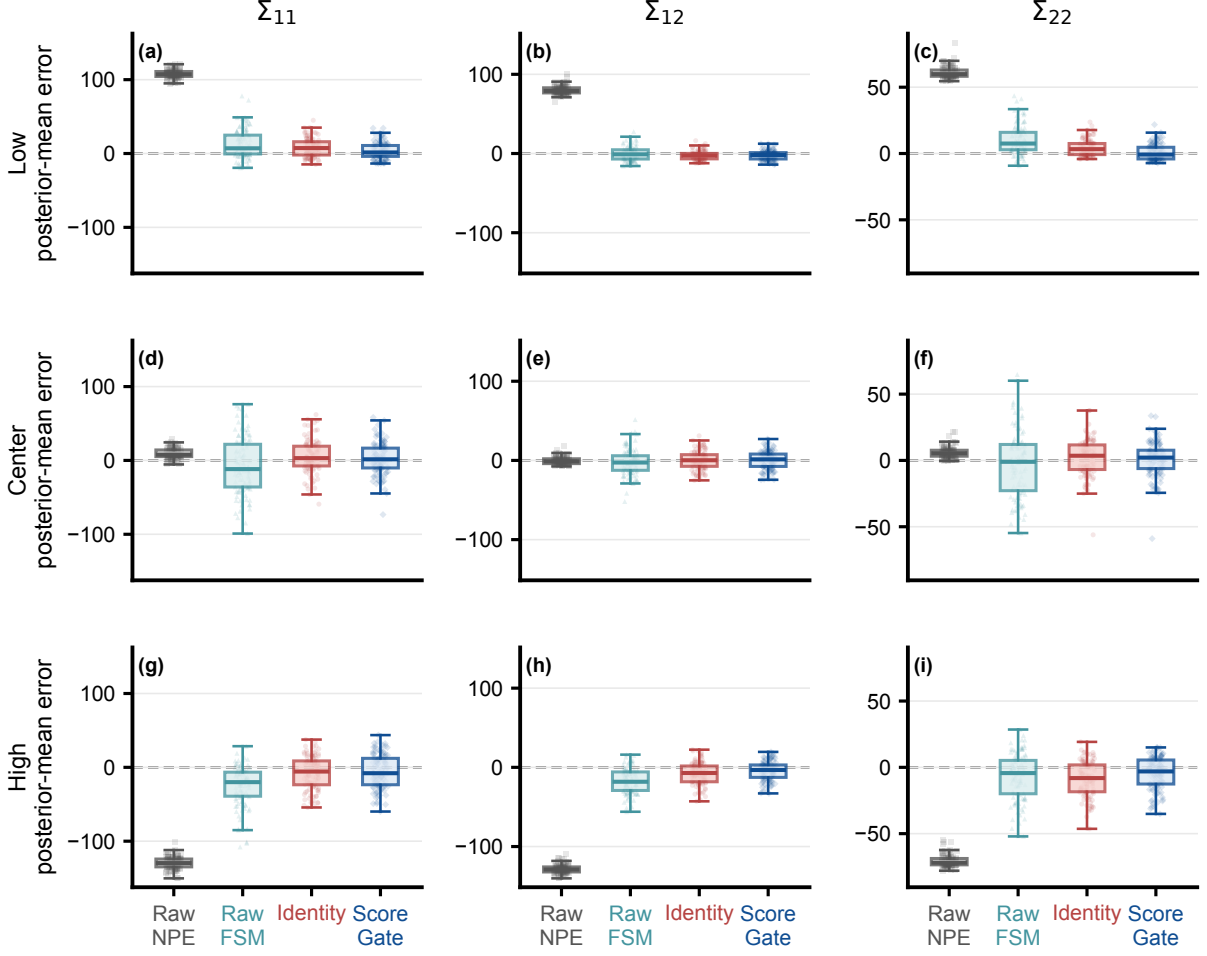


Figure 5: Smith posterior-mean errors for Raw-data NPE, Raw-data FSM, ILSA, and NLSA across 100 paired datasets at Low, Center, and High. Columns are covariance coordinates and rows are generating values. Boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range, and faint points are individual datasets. Dashed lines mark zero error. The three parameter values share vertical limits within each coordinate.

I Details of Models 1 and 2

I.1 A common mixture-score identity

For Models 1 and 2, write $u = \text{logit}(\pi)$, where $\text{expit}(u) = (1 + e^{-u})^{-1}$ denotes the logistic function so that $\pi = \text{expit}(u)$, and let $f_{0,k}$ and $f_{1,k}$ denote the inactive and active densities of block k . With the full block log-likelihood ratio $L_k(y_k) = \log\{f_{1,k}(y_k)/f_{0,k}(y_k)\}$,

$$p(y_k | u) = f_{0,k}(y_k)\{1 - \pi + \pi \exp(L_k)\}, \quad \pi = \text{expit}(u). \quad (19)$$

Because $\partial_u \pi = \pi(1 - \pi)$, direct differentiation gives

$$\frac{\partial}{\partial u} \log p(y_k | u) = \frac{\pi(1 - \pi)\{\exp(L_k) - 1\}}{1 - \pi + \pi \exp(L_k)} \quad (20)$$

$$= \text{expit}(u + L_k) - \text{expit}(u) \equiv h_u(L_k). \quad (21)$$

Thus the exact full-data score and likelihood-grid posterior used only for evaluation are

$$S_{\text{exact}}(Y, u) = \sum_{k=1}^K h_u(L_k), \quad (22)$$

$$p(\pi \mid Y) \propto \mathbb{K}_{[\pi_{\min}, \pi_{\max}]}(\pi) \prod_{k=1}^K \{1 - \pi + \pi \exp(L_k)\}. \quad (23)$$

The omitted factor $\prod_k f_{0,k}(y_k)$ is constant in π . Equation 21 is a score with respect to u , not with respect to π .

At an anchor a , a local likelihood ratio ℓ produces the local score contribution $s = h_a(\ell) = \text{expit}(a + \ell) - \text{expit}(a)$. Writing $\pi_a = \text{expit}(a)$, this link is strictly increasing and has the exact inverse

$$h_a^{-1}(s) = \text{logit}(\pi_a + s) - a, \quad -\pi_a < s < 1 - \pi_a. \quad (24)$$

This inverse is not supplied to the network. Throughout this appendix, the local map follows the stacked-vector convention

$$\phi_{\eta,g}(s, a) = \begin{bmatrix} s \\ m_{\eta,g}(s, a) \end{bmatrix},$$

where stacking, rather than elementwise multiplication, is intended. ILSA uses the dimension-matched embedding $\phi_{0,g}(s, a) = (s, 0)^\top$, whereas NLSA learns the additional component $m_{\eta,g}(s, a)$ from FSM targets. The stacked local features are summed within each group and block. The inverse only characterizes the information available in the local scores.

I.2 Model 1: mean-shift blocks

Conditionally on an inactive block, $Y_k \sim \mathcal{N}(0, I_m)$; conditionally on an active block, $Y_k \sim \mathcal{N}(\tau \mathbf{1}, I_m)$. The observation-level and full-block log-likelihood ratios are therefore

$$\ell_{kj}^{(1)} = \tau Y_{kj} - \frac{\tau^2}{2}, \quad L_k = \sum_{j=1}^m \ell_{kj}^{(1)} = \tau \sum_{j=1}^m Y_{kj} - \frac{m\tau^2}{2}. \quad (25)$$

The architecture receives $s_{kj} = h_a(\ell_{kj}^{(1)})$. If the inverse in Eq. 24 were applied explicitly, summing the recovered local ratios and then applying h_a would give the exact block score $h_a(L_k)$. The experiment deliberately does not hard-code either operation: the two methods differ only in whether a learned nonlinear feature is supplied alongside the identity score before the common within-block pooling and shared block readout. The resulting block-level readout contributions are then summed over $k = 1, \dots, K$.

I.3 Model 2: marginal and pair recovery

An inactive block has distribution $\mathcal{N}(0, I_m)$, whereas an active block has distribution $\mathcal{N}(0, I_m + \tau^2 \mathbf{1}\mathbf{1}^\top)$. Define

$$c_r = -\frac{1}{2} \log(1 + r\tau^2), \quad d_r = \frac{\tau^2}{2(1 + r\tau^2)}, \quad (26)$$

$$T_k = \sum_{i=1}^m Y_{ki}, \quad Q_k = \sum_{i=1}^m Y_{ki}^2. \quad (27)$$

The full, marginal, and within-block pair log-likelihood ratios are

$$L_k = c_m + d_m T_k^2, \quad (28)$$

$$\ell_{ki}^{(1)} = c_1 + d_1 Y_{ki}^2, \quad (29)$$

$$\ell_{k,ij}^{(2)} = c_2 + d_2 (Y_{ki} + Y_{kj})^2, \quad i < j. \quad (30)$$

Let $P = \binom{m}{2}$ and aggregate the local evidence as $A_{1,k} = \sum_i \ell_{ki}^{(1)}$ and $A_{2,k} = \sum_{i<j} \ell_{k,ij}^{(2)}$. The identity

$$\sum_{i<j} (Y_{ki} + Y_{kj})^2 = (m-2)Q_k + T_k^2 \quad (31)$$

implies

$$A_{1,k} = mc_1 + d_1 Q_k, \quad (32)$$

$$A_{2,k} = Pc_2 + d_2 \{(m-2)Q_k + T_k^2\}, \quad (33)$$

$$Q_k = \frac{A_{1,k} - mc_1}{d_1}, \quad (34)$$

$$T_k^2 = \frac{A_{2,k} - Pc_2}{d_2} - (m-2) \frac{A_{1,k} - mc_1}{d_1}. \quad (35)$$

Substitution into $L_k = c_m + d_m T_k^2$ recovers the exact full-block ratio. The actual inputs, however, are the score contributions $h_a(\ell_{ki}^{(1)})$ and $h_a(\ell_{k,ij}^{(2)})$, pooled separately by component type. Both types pass through the same mixture-score link, so the same shared nonlinear map can in principle approximate Eq. 24 for either type. Under the stacked representation, the identity-score sum and learned-feature sum are retained separately for each type; if the learned feature approximates the inverse link, its two group sums recover $A_{1,k}$ and $A_{2,k}$. A shared block readout then combines the two group representations, and its outputs are summed over blocks. As in Model 1, neither the inverse link nor Eq. 35 is coded into Stage 1 or Stage 2; the identities establish what information the local evidence contains and provide exact held-out references.

I.4 Pilots for Models 1 and 2

The Model 1 pilot field is the marginal composite score

$$S_{\text{pilot}}^{(1)}(Y, u) = \sum_{k=1}^K \frac{1}{m} \sum_{j=1}^m h_u(\ell_{kj}^{(1)}). \quad (36)$$

Model 2 gives the marginal and pairwise score groups equal weight,

$$S_{\text{pilot}}^{(2)}(Y, u) = \sum_{k=1}^K \left\{ \frac{1}{m} \sum_i h_u(\ell_{ki}^{(1)}) + \frac{1}{P} \sum_{i < j} h_u(\ell_{k,ij}^{(2)}) \right\}. \quad (37)$$

Equal weighting prevents the pairwise group from dominating merely because $P \gg m$. Model 1 evaluates Eq. 36 at 201 probabilities evenly spaced over $[0.05, 0.70]$ and maps them to u . Model 2 evaluates Eq. 37 on 201 points over the interval

$$[\text{logit}(0.05), \text{logit}(0.70)].$$

For each dataset, the implementation uses trapezoidal integration of the score function to obtain a potential, linearly interpolates all sign-change roots, includes both support boundaries as candidates, and selects the global constrained maximizer of the potential. Both methods share the resulting $\hat{u}(Y)$. These pilots use only local composite scores: neither the full ratio, the recovery identity, the exact likelihood, nor the generating π enters their construction.

J Details of Model 3

J.1 Simulation design

Each simulated dataset consists of 47 independent annual-maxima fields at the locations of 79 Swiss stations. Inference targets the storm-shape covariance $\Sigma = (\Sigma_{11}, \Sigma_{12}, \Sigma_{22})$. We evaluate three fixed generating covariances:

$$\begin{aligned} \text{Low} &= (234.1, -10.1, 130.1), \\ \text{Center} &= (332.2, 70.4, 184.6), \\ \text{High} &= (471.3, 199.1, 262.0), \end{aligned}$$

where the entries are ordered as $(\Sigma_{11}, \Sigma_{12}, \Sigma_{22})$. Center is the rainfall-reference covariance, while Low and High are prespecified departures on opposite sides of this reference. Smith uses one Stage-1 training run and 100 paired evaluation datasets under each regime, with all methods evaluated on the same simulated datasets.

The station coordinates and rainfall-reference covariance are based on the **rainfall** dataset distributed with **SpatialExtremes** v2.1-0 [Ribatet, 2024]. The dataset contains summer maximum daily rainfall from 1962–2008 at 79 Swiss stations. We obtain the reference covariance and fixed GEV marginal parameters by fitting the Smith model to these observations using maximum pairwise likelihood. We then simulate Smith max-stable fields using **rmaxstab** and transform the unit-Fréchet simulations to the fixed GEV margins using **frech2gev**. **SpatialExtremes** is distributed under GPL version 2 or later.

J.2 Covariance parameterization

The implementation uses a bounded design coordinate $\xi \in [0.15, 0.85]^3$ that maps one-to-one to the reported covariance through fixed monotone maps around the rainfall reference $(\Sigma_{11,0}, \Sigma_{12,0}, \Sigma_{22,0}) = (332.1527, 70.3982, 184.6266)$. Let $\sigma_{x,0} = \sqrt{\Sigma_{11,0}}$, $\sigma_{y,0} = \sqrt{\Sigma_{22,0}}$,

and $\varrho_0 = \Sigma_{12,0}/(\sigma_{x,0}\sigma_{y,0})$. Then

$$\sigma_x(\xi) = \sigma_{x,0} \exp\{0.70(\xi_1 - 0.5)\}, \quad (38)$$

$$\sigma_y(\xi) = \sigma_{y,0} \exp\{0.70(\xi_2 - 0.5)\}, \quad (39)$$

$$\varrho(\xi) = \tanh\{\operatorname{arctanh}(\varrho_0) + 1.40(\xi_3 - 0.5)\}, \quad (40)$$

$$\Sigma(\xi) = \begin{pmatrix} \sigma_x^2 & \varrho\sigma_x\sigma_y \\ \varrho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix}. \quad (41)$$

The three maps equal the rainfall reference when $\xi = (0.5, 0.5, 0.5)$ and preserve positive definiteness. The Low, Center, and High test designs use $\xi = (0.25, 0.25, 0.25)$, $(0.50, 0.50, 0.50)$, and $(0.75, 0.75, 0.75)$, respectively. All anchors, scores, and pilot optimization use the unconstrained working coordinate

$$u_r = \operatorname{logit}\left(\frac{\xi_r - 0.15}{0.70}\right), \quad \xi_r = 0.15 + 0.70 \operatorname{expit}(u_r). \quad (42)$$

Thus the Smith instance of the generic inverse map is $r^{-1}(u) = \Sigma(\xi(u))$. We fix the remaining seven GEV marginal parameters at the rainfall reference and infer only the three coordinates controlling $\Sigma(\xi)$. We train the NPE in ξ and map its draws one-to-one to Σ for every reported posterior summary.

J.3 Exact bivariate score contributions

For sites i and j , let $h_{ij} \in \mathbb{R}^2$ be their spatial displacement and define $a_{ij} = \sqrt{h_{ij}^\top \Sigma^{-1} h_{ij}}$. On the unit-Fréchet scale, the Smith bivariate exponent function is

$$V_{ij}(z_i, z_j; \Sigma) = \frac{\Phi(w)}{z_i} + \frac{\Phi(v)}{z_j}, \quad w = \frac{a_{ij}}{2} + \frac{\log(z_j/z_i)}{a_{ij}}, \quad v = a_{ij} - w, \quad (43)$$

where Φ is the standard normal distribution function. It follows that $F_{ij} = \exp(-V_{ij})$ and

$$f_{ij}(z_i, z_j; \Sigma) = \exp(-V_{ij})\{V_i V_j - V_{ij}^{(12)}\}, \quad (44)$$

where subscripts on V denote partial derivatives with respect to z_i and z_j . The implemented pair likelihood applies the fixed generalized extreme-value (GEV) to unit-Fréchet transformation and its Jacobian before evaluating Eq. 44. For year t , the exact local score contribution is

$$s_{t,ij}(u_0) = \nabla_u \log f_{ij}(Y_{ti}, Y_{tj}; \Sigma(\xi(u)))|_{u=u_0} \in \mathbb{R}^3, \quad (45)$$

computed by automatic differentiation of the exact bivariate objective. All 3,081 pairs are partitioned into 20 fixed direction-by-distance groups, formed from four axial-angle groups and five equal-count distance groups. ILSA or NLSA acts on each pairwise score before the mean is taken within its group, and the common readout then aggregates the 47 independent years. We do not know an analytic inverse of Eq. 45, and no exact 79-site likelihood is used.

The covariance implies the Smith extremal coefficient

$$\delta_\Sigma(h) = 2\Phi\left(\frac{1}{2}\sqrt{h^\top \Sigma^{-1} h}\right), \quad (46)$$

which summarizes extremal dependence at spatial displacement h . Evaluation at the three fixed parameter values uses the normalized equal-weight grid approximation in Eq. 56 over the common lag domain.

J.4 Composite-likelihood pilot

For the pilot, we select 25 station pairs once within each of the 20 groups. If $\mathcal{G}_g$ is the full group and $\tilde{\mathcal{G}}_g$ its selected subset, the rough composite objective is

$$Q(u; Y) = \sum_{t=1}^{47} \sum_{g=1}^{20} \frac{|\mathcal{G}_g|}{|\tilde{\mathcal{G}}_g|} \sum_{(i,j) \in \tilde{\mathcal{G}}_g} \log f_{ij}(Y_{ti}, Y_{tj}; \Sigma(\xi(u))). \quad (47)$$

The population-count weight makes the 500-pair subset approximate the all-pair composite objective. Optimization is deterministic and constrained to $[-6, 6]^3$ in u . Fifteen starts comprise the origin, the six signed coordinate starts at radius 2, and eight corner starts at radius $2/\sqrt{3}$. We refine the three starts with the largest initial objective for 12 Newton iterations. Each Hessian solve uses ridge 10^{-3} , caps the step norm at 1, and chooses the best finite candidate from six halvings of the proposal plus the current point. The final pilot takes the refined candidate with the largest objective. A projected-score residual threshold of 10^{-3} audits convergence. ILSA and NLSA use the same pilot, which the procedure computes without the generating parameter or a full likelihood.

Table 4 in the main text reports covariance-scale MSE for 100 paired datasets at each parameter value. We include the all-pair maximum pairwise-likelihood estimate (MPLE) as a point-estimation benchmark. We compute its normalized extremal ISE by plugging each MPLE covariance estimate into Eq. 46 on the same lag grid used for the neural methods. MPLE has no posterior output; its numerical uncertainty interval in the representative figure below serves only as an illustration.

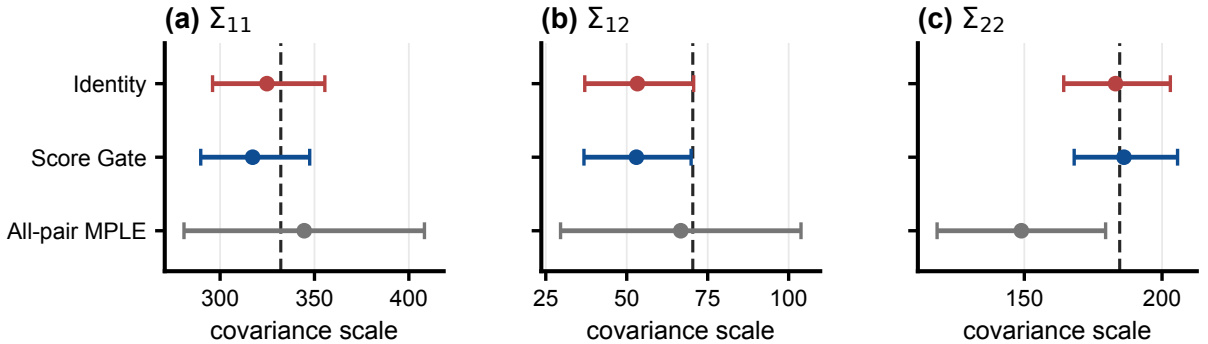


Figure 6: Representative Model 3 uncertainty for test dataset 0 at the prespecified Center parameter value. Points are posterior means for ILSA and NLSA and the all-pair MPLE; horizontal segments are central 90% credible intervals for NPE and the corresponding 90% MPLE interval. Dashed vertical lines mark the generating covariance coordinates. This is an illustrative paired dataset, not a replacement for the 100-dataset MSE summaries.

K Training and evaluation

Within each model, ILSA and NLSA use the same simulated data, architecture, optimization settings, checkpoint rule, data-only pilot, and Stage-2 posterior estimator; only

the pre-pooling map differs. Its initialization makes NLSA reproduce ILSA under the corresponding model-specific preprocessing, which we verify numerically before training.

For Stage 1, all models use learning rate 10^{-4} , weight decay 10^{-3} , gradient clipping at 5, and EMA decay 0.995. Models 1–2 use validation-best checkpoints, whereas Smith uses the fixed final EMA checkpoint. The shared readout has width 64 and depth 2, and the NLSA has hidden width 16. For Stage 2, all models use an 8-component MDN-NPE with hidden width 64, batch size 256, and learning rate 5×10^{-4} . Table 7 summarizes the model-specific settings.

Table 7: Model-specific training settings. All models share the remaining optimization and architecture settings described in the text.

	Model 1	Model 2	Smith
Stage-1 train / validation	40k / 8k	40k / 8k	10k / 8k
Stage-1 updates / batch	20k / 512	20k / 512	10k / 128
Proposal s.d. σ_q	0.20	0.15	0.20 per coordinate
Stage-2 simulations	50k	50k	10k

Models 1–2 use five Stage-1 training repetitions. Stage-1 exact-score evaluation uses 5,000 held-out datasets at each of four generating values for Model 1 and six generating values for Model 2. Stage-2 posterior evaluation uses 10 paired datasets at each of six generating values per repetition. Smith uses one Stage-1 training run and 100 paired evaluation datasets at each of three fixed parameter values.

L FSM target and evaluation measures

L.1 Population target of anchored FSM

Let $q(\tilde{u} \mid u_0)$ be the Gaussian proposal used in Stage 1 and define the proposal-smoothed likelihood

$$m_q(Y \mid u_0) = \int p(Y \mid \tilde{u}) q(\tilde{u} \mid u_0) d\tilde{u}. \quad (48)$$

For a Gaussian with covariance Σ_q , $\nabla_{u_0} \log q(\tilde{u} \mid u_0) = \Sigma_q^{-1}(\tilde{u} - u_0)$. Differentiating under the integral gives

$$\nabla_{u_0} \log m_q(Y \mid u_0) = \frac{\int p(Y \mid \tilde{u}) q(\tilde{u} \mid u_0) \Sigma_q^{-1}(\tilde{u} - u_0) d\tilde{u}}{m_q(Y \mid u_0)} \quad (49)$$

$$= \mathbb{E}[\Sigma_q^{-1}(\tilde{u} - u_0) \mid Y, u_0]. \quad (50)$$

Hence an unrestricted regressor receiving the full data learns the score of the smoothed likelihood at population level. In the actual Stage-1 architecture the regressor sees only the local-score representation. If $U_{u_0}(Y)$ denotes the complete pre-pooling local score collection evaluated at anchor u_0 , its unrestricted Bayes target is instead

$$\mathbb{E}[\Sigma_q^{-1}(\tilde{u} - u_0) \mid U_{u_0}(Y), u_0] = \mathbb{E}[\nabla_{u_0} \log m_q(Y \mid u_0) \mid U_{u_0}(Y), u_0]. \quad (51)$$

Equivalently, by Fisher’s identity this is the conditional expectation of the Fisher score $\nabla_{\tilde{u}} \log p(Y \mid \tilde{u})$ under the local posterior, conditional on the available local scores. As

$\Sigma_q \rightarrow 0$, this target approaches the conditional projection of the original Fisher score at u_0 onto the full local score collection. Finite proposal width and subsequent transform–pool compression therefore introduce distinct approximations.

L.2 Smith covariance and dependence measures

Let N_{test} be the number of test datasets and M the number of posterior draws used for each dataset. For a neural method, let $\xi_i^{(1)}, \dots, \xi_i^{(M)} \sim q_\lambda\{\xi \mid c(Y_i)\}$ be draws in the bounded Smith design coordinate for test dataset $i = 1, \dots, N_{\text{test}}$. The reported covariance estimate averages on the covariance scale,

$$\widehat{\Sigma}_i = \frac{1}{M} \sum_{m=1}^M \Sigma(\xi_i^{(m)}) \approx \mathbb{E}_{q_\lambda(\xi|c(Y_i))}[\Sigma(\xi)], \quad (52)$$

not $\Sigma\{M^{-1} \sum_m \xi_i^{(m)}\}$. If $\Sigma^\star$ is the generating covariance, we evaluate coordinate $a \in \{11, 12, 22\}$ by

$$\text{MSE}_a = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left(\widehat{\Sigma}_{i,a} - \Sigma_a^\star \right)^2. \quad (53)$$

The Smith extremal coefficient at spatial displacement h is

$$\delta_\Sigma(h) = 2\Phi\left\{ \frac{\sqrt{h^\top \Sigma^{-1} h}}{2} \right\}. \quad (54)$$

The neural estimate averages after this nonlinear transformation,

$$\widehat{\delta}_i(h) = \frac{1}{M} \sum_{m=1}^M \delta_{\Sigma(\xi_i^{(m)})}(h) \approx \mathbb{E}_{q_\lambda(\xi|c(Y_i))}[\delta_{\Sigma(\xi)}(h)]. \quad (55)$$

For MPLE, plugging in its point estimate yields the surface. On the uniformly weighted displacement grid $\mathcal{H}$, the per-dataset normalized grid ISE is

$$\text{ISE}_i = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \left\{ \widehat{\delta}_i(h) - \delta_{\Sigma^\star}(h) \right\}^2. \quad (56)$$

L.3 Aggregation of evaluation results

For Models 1 and 2, we first compute each metric within a fixed generating value and then average with equal weight over the prespecified values. Stage-1 standardization uses the exact-score variance within that value. For Smith, we average covariance errors over the paired datasets within each generating parameter value. Normalized extremal ISE also averages uniformly over the common displacement grid in Eq. 56. We never pool metrics across differently scaled covariance coordinates.

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